PROPOSED DUAL-OCCUPANCY
18 CARRAMAR CRESCENT, MIRANDA NSW

SAFETY IN DESIGN

THERE ARE INHERENT RISKS WITH CONSTRUCTING, MAINTAINING, OPERATING, DEMOLISHING, DISMANTLING AND DISPOSING THIS DESIGN THAT ARE TYPICAL OF SIMILAR DESIGNS. AS FAR AS IS REASONABLY PRACTICABLE RISKS HAVE BEEN ELIMINATED OR MINIMISED THROUGH THE DESIGN PROCESS. HAZARD CONTROLS MUST STILL BE IMPLEMENTED BY THE CONTRACTOR, OWNER OR OPERATOR TO ENSURE THE SAFETY OF WORKERS. . QDE DO NOT CONSIDER THAT THERE ARE ANY UNIQUE RISKS ASSOCIATED WITH THE DESIGN OF THIS PROJECT.

DRAWING STATUS

PRELIMINARY PRELIMINARY DRAWINGS ARE NOT TO BE USED FOR TENDER OR CONSTRUCTION PURPOSES.
TENDER TENDER DRAWINGS ARE NOT TO BE USED FOR CONSTRUCTION PURPOSES AND ARE INTENDED FOR AN EXTENT OF WORKS. ALL OTHER CONSULTANT DRAWINGS AND CONTRACT DOCUMENTS SHOULD BE READ IN CONJUNCTION WITH THESE DOCUMENTS TO DETERMINE THE FULL EXTENT OF WORKS.
CONSTRUCTION CERTIFICATE CONSTRUCTION CERTIFICATE DRAWINGS ARE NOT TO BE USED FOR TENDER OR CONSTRUCTION
CONSTRUCTION CONSTRUCTION DRAWINGS CAN BE USED FOR CONSTRUCTION PURPOSES AND AS REQUIRED FOR THE CREATION OF FABRICATION/WORKSHOP DRAWINGS.

GENERAL

1. THESE DRAWINGS SHALL BE READ IN CONJUNCTION WITH ALL ARCHITECTURAL AND OTHER CONSULTANTS' DRAWINGS AND SPECIFICATION AND WITH SUCH OTHER WRITTEN INSTRUCTION AS MAY BE ISSUED DURING THE COURSE OF THE CONTRACT. ANY DISCREPANCY SHALL BE REFERRED TO THE SUPERINTENDENT BEFORE PROCEEDING WITH THE WORK.
2. ALL MATERIALS AND WORKMANSHIP SHALL BE IN ACCORDANCE WITH THE RELEVANT AND CURRENT AUSTRALIAN STANDARDS AND THE NATIONAL CONSTRUCTION CODE SERIES EXCEPT WHERE VARIED BY THE PROJECT SPECIFICATION.
3. THESE DRAWINGS ARE DIAGRAMMATIC AND SHOW THE GENERAL ARRANGEMENT. ALL DIMENSIONS SHOWN SHALL BE VERIFIED BY THE BUILDER ON SITE. ENGINEERS' DRAWINGS SHALL NOT BE SCALED FOR DIMENSIONS.
4. DURING CONSTRUCTION THE STRUCTURE SHALL BE MAINTAINED IN A STABLE CONDITION AND NO PART SHALL BE OVERSTRESSED. TEMPORARY BRACING SHALL BE PROVIDED BY THE BUILDER TO KEEP THE WORKS AND EXCAVATIONS STABLE AT ALL TIMES.
5. THE BUILDER SHALL GIVE 48 HOURS NOTICE FOR ALL ENGINEERING INSPECTIONS.
6. THE BUILDER SHALL LOCATE ALL EXISTING SERVICES PRIOR TO COMMENCEMENT OF CONSTRUCTION. ENGINEER TO BE NOTIFIED.
7. UNLESS NOTED OTHERWISE ALL LEVELS ARE IN METRES AND ALL DIMENSIONS ARE IN MILLIMETRES.
8. ALL PROPRIETY ITEMS TO BE INSTALLED IN ACCORDANCE WITH MANUFACTURES SPECIFICATIONS & DETAILS.

STRUCTURAL DESIGN CRITERIA

PERMANENT & IMPOSED LOADING (AS1170.1)

FLOOR USAGE	IMPOSED LOAD (kPa)	SUPERIMPOSED PERMANENT LOAD(kPa)
RESIDENTIAL	1.5	1.0
BALCONY	2.0	1.0
NON TRAFFICABLE ROOF	0.25	0.95/0.5(TILE/METAL)
STAIR	2.0	-

WIND DESIGN CRITERIA (AS1170.2)

WIND REGION:	A2
IMPORTANCE LEVEL (BCA):	2
ANNUAL PROBABILITY OF EXCEEDENCE	1 in 500
TERRAIN CATEGORY:	III
SHIELDING CLASSIFICATION:	Ms = 1.0
TOPOGRAPHIC CLASSIFICATION:	Mt = 1.0
REGIONAL WIND SPEED:	Vu = 45m/s

FOUNDATION DESIGN CRITERIA

FOUNDATION MATERIAL	DENSE SAND
ALLOWABLE BEARING PRESSURE	ASSUMED 150 KPA
GEOTECHNICAL INVESTIGATIONS REPORT No	-

PROJECT SPECIFICATION	
SITE CLASSIFICATION	ASSUMED CLASS S
FOUNDATION MATERIAL	ASSUMED DENSE SAND WITH ALLOWABLE BEARING CAPACITY OF 150 kPa
TYPE OF CONSTRUCTION	DOUBLE BRICK WALL ON GF & BRICK VENEER FIRST FLOOR
SALINITY AFFECTED	BUILDER TO CONFIRM
WIND CLASSIFICATION	N2
THE GEOTECHNICAL INVESTIGATION TO BE CONDUCTED BEFORE CONSTRUCTION. WE TAKE NO RESPONSIBILITY FOR VARIATIONS THAT MAY OCCUR DUE TO VARIATIONS IN SITE CONDITIONS	

CONCRETE DESIGN

CEMENT TYPE	"GP"
LIMIT DRYING SHRINKAGE AT 56 DAYS	800 MICRONS
SULPHATE RESISTANCE CONCRETE BELOW GROUND	YES

ELEMENT	STRENGTH	COVER EXT	COVER INT
FOOTING	32Mpa	50	-
SUSPENDED SLABS	40MPa	40	25
BEAMS	40MPa	40	25
COLUMNS	40MPa	40	30
WALLS	40MPa	40	30
STAIRS	32Mpa	40	25

CONCRETE

1. ALL WORKMANSHIP AND MATERIALS SHALL BE IN ACCORDANCE WITH AS 3600 AND AS 3610 CURRENT EDITIONS WITH AMENDMENTS, EXCEPT WHERE VARIED BY THE CONTRACT DOCUMENTS.
2. PROJECT CONTROL TESTING SHALL BE CARRIED OUT IN ACCORDANCE WITH AS 3600.
3. CONCRETE TO BE PROPORTIONED TO LIMIT DRYING SHRINKAGE TO SATISFY PROJECT INFORMATION TABLE.
4. NO ADMIXTURES SHALL BE USED IN CONCRETE UNLESS APPROVED IN WRITING.
5. CLEAR CONCRETE COVER TO ALL REINFORCEMENT SHALL BE AS NOTED IN THE PROJECT INFORMATION TABLE UNLESS SHOWN OTHERWISE.
6. WHERE CONCRETE IS POURED ON A VAPOUR PROOF MEMBRANE 0.2mm MINIMUM THICKNESS, THE COVER TO CONCRETE CAST AGAINST GROUND MAY BE REDUCED BY 10mm.
7. CONCRETE SIZES SHOWN DO NOT INCLUDE THICKNESS OF APPLIED FINISHES. NO FINISH WHICH DECREASES COVER IS ALLOWED WITHOUT THE WRITTEN APPROVAL OF THE ENGINEER.
8. DEPTHS OF BEAMS ARE GIVEN FIRST AND INCLUDE SLAB THICKNESS.
9. FOR CHAMFERS DRIP GROOVES, REGLETS, ETC. REFER TO ARCHITECT'S DETAILS. MAINTAIN COVER TO REINFORCEMENT AT THESE DETAILS.
10. NO HOLES, CHASES, BLOCKOUTS, DUCTS OR EMBEDMENT OF PIPES OTHER THAN THOSE SHOWN ON THE STRUCTURAL DRAWINGS SHALL BE MADE IN CONCRETE MEMBERS WITHOUT THE PRIOR WRITTEN APPROVAL OF THE ENGINEER.
11. CONSTRUCTION JOINTS WHERE NOT SHOWN SHALL BE LOCATED TO THE APPROVAL OF THE ENGINEER.
12. ALL CONCRETE COLUMNS GREATER THAN 1.2 METRES IN HEIGHT SHALL BE POURED A MINIMUM OF 4 HOURS PRIOR TO SLAB OR BEAM OVER.
13. CONCRETE SHALL BE MECHANICALLY VIBRATED TO ACHIEVE A DENSE HOMOGENEOUS MASS COMPLETELY FILLING THE FORMWORK THOROUGHLY EMBEDDING THE REINFORCEMENT AND FREE OF STONE POCKETS OR VOIDS. ALL CONCRETE INCLUDING SLABS ON GROUND AND FOOTINGS SHALL BE COMPACTED WITH MECHANICAL VIBRATORS.
14. CURING OF ALL CONCRETE IS TO BE ACHIEVED BY KEEPING SURFACES CONTINUOUSLY WET AS PER AS 3600. INTERNAL SLABS SHOULD BE CURED FOR A MINIMUM OF 3 DAYS AND EXTERNAL SLABS FOR A MINIMUM OF 7 DAYS. CURING COMPOUNDS MAY BE USED BUT MUST BE COMPATIBLE WITH THE PROPOSED FLOOR FINISHED & BE INSTALLED IN ACCORDANCE WITH MANUFACTURERS SPECIFICATIONS. POLYTHENE SHEETING OR WET HESSIAN MAY BE USED IF PROTECTED FROM WIND AND TRAFFIC.
15. CONSTRUCTION SUPPORT PROPPING IS TO BE LEFT IN PLACE WHERE NEEDED TO AVOID OVERSTRESSING THE STRUCTURE DUE TO CONSTRUCTION LOADING. NO BRICKWORK OR PARTITION WALLS ARE TO BE CONSTRUCTED ON SUSPENDED LEVELS UNTIL SEVEN DAYS AFTER PROPPING HAS BEEN REMOVED AND THE SLAB PRE-LOADED WITH THE BRICKS OR UNITS TO BE USED IN THE WALL.
16. REPAIRS TO CONCRETE SHALL NOT BE ATTEMPTED WITHOUT THE PERMISSION OF THE ENGINEER.
17. CAST-IN FIXINGS, BOLTS ETC. SHALL NOT BE ALTERED WITHOUT THE PERMISSION OF THE ENGINEER.
18. THE CONCRETE SHALL BE PLACED IN SUCH A MANNER AS TO AVOID SEGREGATION OR LOSS OF MATERIALS. MAXIMUM FALL OF CONCRETE = 1500mm OR USE ENCLOSED CHUTES OR SIMILAR.
19. SULPHATE RESISTANT CONCRETE TO BE USED IN ALL FOOTINGS, PILES AND PILE CAPS IF REQUIRED BY THE PROJECT DESIGN INFORMATION.
20. IF AMBIENT TEMPERATURES ARE LESS THAN 5°C OR GREATER THAN 35°C DO NOT MIX CONCRETE WITHOUT TAKING PROVISIONS TO ENSURE THE CONCRETE IS DELIVERED WITHIN THIS TEMPERATURE RANGE.

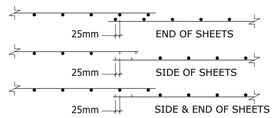
MASONRY DESIGN

LOCATION	RESISTANCE GRADE	MIN MORTAR CLASS	DURABILITY CLASS OF BUILT IN COMPONENTS
INTERNAL	PROTECTED	M3	R1
EXTERNAL	PROTECTED	M3	R1
BELOW GROUND	GP	M3	R3
EXT(MARINE)	GP	M3	R3
EXT(SEVERE MARINE)	EXPOSURE	M4	R4
EXT(INDUSTRIAL)	EXPOSURE	M4	R4

- THE ABOVE TABLE IS FOR BOTH BRICKWORK & BLOCKWORK EXTERNAL
- BELOW GROUND DENOTES BELOW DAMP PROOF COURSE OR IN CONTACT WITH SOIL
- CORE FILL BLOCKS WITH 20MPa GROUT - 10mm MAX AGGREGATE SIZE

REINFORCEMENT

1. REINFORCEMENT IS REPRESENTED DIAGRAMMATICALLY AND NOT NECESSARILY IN TRUE PROJECTION.
2. SPLICES IN REINFORCEMENT SHALL BE MADE ONLY IN POSITIONS SHOWN OR 0THERWISE APPROVED IN WRITING BY THE ENGINEER. LAPS SHALL BE IN ACCORDANCE WITH THE LAP TABLE.
3. WELDING OF REINFORCEMENT SHALL NOT BE PERMITTED UNLESS SHOWN ON THE STRUCTURAL DRAWINGS OR APPROVED BY THE ENGINEER.
4. FABRIC SHALL BE LAPPED 2 TRANSVERSE WIRES PLUS 50mm. WHERE FABRIC LAPS, SHEETS TO HAVE MAXIMUM 2 LAYERS AT ANY POINT, CUT BACK FABRIC AT CORNERS AS REQUIRED.



5. BUNDLED BARS SHALL BE TIED TOGETHER AT 30 BAR DIAMETER CENTERS WITH 3 WRAPS OF THE WIRE.
6. FIRE RATING MESH F41 SHALL BE GALVANISED, HAVE 20mm COVER AND BE TIED USING GALVANISED TIE WIRE
7. WHERE TRANSVERSE TIE BARS ARE NOT SHOWN PROVIDE N12-400 LAPPED AS REQUIRED.
8. JOGGLES TO BARS SHALL COMPRISE A LENGTH OF 12 BAR DIAMETERS BETWEEN BEGINNING AND END OF AN OFFSET OF 1 BAR DIAMETER.
9. ALL REINFORCEMENT SHALL BE FIRMLY SUPPORTED ON MILD STEEL PLASTIC TIPPED CHAIRS, PLASTIC CHAIRS OR CONCRETE CHAIRS AT NOT GREATER THAN 1 METRE CENTRES BOTH WAYS, AND 800 EACH WAY FOR FABRIC. WHEN POURED ON GROUND PROVIDE PLATES UNDER ALL BAR CHAIRS. PLASTIC TIPPED STEEL CHAIRS SHALL NOT BE USED ON EXPOSED FACES.
10. REINFORCEMENT WITHIN FLOOR OR WALL ELEMENT OF A WET AREA SHALL BE BONDED TO THE EARTHING SYSTEM IF CONDUCTIVE PIPING EXISTS WITHIN THE ELEMENT. IT SHALL BE BONDED IN ACCORDANCE WITH AS 3000.
11. REINFORCEMENT SYMBOLS:
    - R- GRADE 250 R HOT ROLLED PLAIN BARS TO AS 4671
    - S-GRADE 230 S HOT ROLLED DEFORMED BARS TO AS 4671
    - N- DEFORMED BAR NORMAL DUCTILITY TO AS/NZS 4671 GRADE D 500N
    - L- DEFORMED BAR LOW DUCTILITY TO AS/NZS 4671 GRADE D500L
    - RN-RECTANGULAR WIRE MESH NORMAL DUCTILITY TO AS/NZS 467
    - RL-RECTANGULAR WIRE MESH LOW DUCTILITY TO AS 4671
    - SN-SQUARE WIRE MESH NORMAL DUCTILITY TO AS 4671
    - SL-SQUARE WIRE MESH LOW DUCTILITY TO AS/NZS 4671
12. THE LAP TABLES ARE BASED ON 32MPa CONCRETE WITH 25mm MINIMUM COVER GENERALLY & 30mm COVER FOR BEAMS & COLUMNS.

PILES & SLABS/BEAMS LESS THAN 300mm THICKTENSION LAP LENGTH FOR DEFORMED BARS IN WALLS, COLUMNS,						
N12	N16	N20	N24	N28	N32	N36
500	750	1000	1200	1500	1800	2100

WHERE 300mm OR MORE CONCRETE BELOW BAR TTENSION LAP LENGTH FOR DEFORMED BARS IN SLABS & BEAMS						
N12	N16	N20	N24	N28	N32	N36
650	950	1300	1600	1950	2300	2700

13. INCREASE TABULATED VALUES BY NOMINATED % IN THE FOLLOWING CASES . 25MPa CONCRETE 15% . SLIP FORM CONSTRUCTION INCREASE BY 30% . LIGHTWEIGHT CONCRETE 30% . 3 BUNDLE BARS 20% . 4 BUNDLE BARS 33% . WHERE LAPPING DIFFERENT SIZED BARS, USE LAP LENTH APPLICABLE TO LARGER SIZED BARS
14. DO NOT BEND BARS WITHOUT PRIOR APPROVAL FROM ENGINEER. FOLLOW AS 3600 FOR PROCEDURES OF BENDING BARS. DO NOT USE HEAT TO BEND BARS AND BENDING OF GALVANIZED BARS IS NOT PERMITTED.

TIMBER

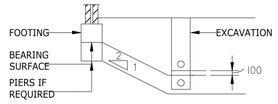
1. ALL TIMBER DESIGN AND CONSTRUCTION SHALL BE TO AS 1720 OR AS 1684 WHERE APPROPRIATE. ALL SOFTWOOD SHALL BE GRADE MGP10 UNLESS NOTED OTHERWISE. ALL HARDWOOD SHALL BE MINIMUM GRADE F11 U.N.O.
2. DURABILITY GRADE OF ALL TIMBERS TO COMPLY WITH THE PROJECT INFORMATION TABLE.
3. ALL BOLTS IN TIMBER CONSTRUCTION SHALL BE MINIMUM M16 UNLESS NOTED OTHERWISE AND SHALL BE GALVANISED. BOLTS SHALL BE RE-TIGHTENED AT THE END OF THE MAINTENANCE PERIOD. ALL TRUSSES AND RAFTERS SHALL BE FIXED TO TOP PLATE WITH METAL PLATE CONNECTORS

TIMBER DURABILITY

LOCATION	HAZARD CLASS (SOFT WOODS)	DURABILITY CLASS (HARD WOODS)
INTERNAL ABOVE GROUND	H2	CLASS 1 OR 2
EXTERNAL ABOVE GROUND	H3	CLASS 1 OR 2
BELOW GROUND	H5	CLASS 1

FOUNDATIONS & EARTHWORKS

1. THIS FOUNDATION MATERIAL SHALL BE UNIFORM AND BE APPROVED BY A GEOTECHNICAL ENGINEER FOR THE ALLOWABLE BEARING PRESSURE BEFORE PLACING REINFORCEMENT OR CONCRETE. PIERS OR BLOCKDOWNS SHALL BE USED TO ACHIEVE UNIFORM BEARING WHERE NECESSARY.
2. THIS SITE CLASS HAS BEEN DESIGNED IN ACCORDANCE WITH AS 2870 (WHERE APPLICABLE). CONFIRMATION OF SITE CLASSIFICATION IS REQUIRED BY A GEOTECHNICAL ENGINEER PRIOR TO CONSTRUCTION COMMENCING ON SITE.
3. FOOTINGS SHALL BE CONCRETED ON THE DAY OF APPROVAL UNLESS PERMISSION IS GIVEN OTHERWISE.
4. FOOTINGS SHALL BE LOCATED CENTRALLY UNDER WALLS AND COLUMNS UNLESS NOTED OTHERWISE.
5. FOOTING LEVELS WHERE SHOWN ARE ESTIMATES ONLY AND WILL BE ESTABLISHED DURING SITE INSPECTION OF WORK IN PROGRESS.
6. BACKFILL TO RETAINING WALLS SHALL BE FREE DRAINING GRANULAR MATERIAL. PROVIDE SUBSOIL DRAIN AT BASE OF WALL. DO NOT BACKFILL UNTIL 14 DAYS AFTER CORE FILLING, OR IF APPLICABLE, AFTER RESTRAINING SLAB OVER HAS BEEN POURED AND CURED FOR 7 DAYS. BACKFILL SHALL BE COMPACTED TO 98% STANDARD MAXIMUM DRY DENSITY AT OPTIMUM MOISTURE CONTENT ±2%.
7. PROVIDE ADEQUATE BATTERS OR SHORING TO THE SIDE OF ALL EXCAVATIONS. REFER TO GEOTECHNICAL ENGINEER FOR RELEVANT ADVICE.



EXCAVATION IN ROCK

1. THE CONTRACTOR SHALL SUBMIT HIS PROPOSAL FOR EXCAVATION TO THE SUPERINTENDENT PRIOR TO COMMENCEMENT OF WORK.
2. THE METHOD SHALL COMPLY WITH ALL THE REQUIREMENTS OF THE DA CONDITIONS, PARTICULARLY IN RELATION TO NOISE LEVELS, VIBRATION AND WORK HOURS.
3. VIBRATIONS SHALL NOT EXCEED THAT SPECIFIED IN THE GEOTECHNICAL REPORT AND ALARMS SHALL BE INSTALLED TO MONITOR THESE VIBRATIONS. U.N.O.
4. ALL THE REQUIREMENTS OF THE GEOTECHNICAL REPORT FOR THE SITE SHALL BE OBSERVED AND MET.

POST-INSTALLED REINFORCEMENT AND ANCHORS

1. WHERE SHOWN ON THE DRAWINGS REINFORCEMENT BARS AND BOLTING ANCHORS SHALL BE CHEMICALLY ANCHORED INTO EXISTING CONCRETE AS DESCRIBED BELOW.
2. PERCUSSION DRILL (CORING NOT PERMITTED) EACH HOLE TO THE CORRECT DIAMETER AND MINIMUM DEPTH FOR THE PARTICULAR SIZE REINFORCING BAR OR ANCHOR AS TABULATED BELOW, UNLESS SHOWN OTHERWISE ON THE DRAWINGS.

BAR SIZE	HOLE DIAMETER (mm)	MINIMUM HOLE DEPTH (mm)
12	16	250
16	20	350
20	26	420
24	30	550

ANCHOR SIZE	HOLE DIAMETER (mm)	MINIMUM HOLE DEPTH (mm)
M12	14	120
M16	18	160
M20	24	200
M24	28	240

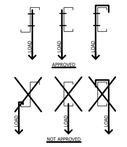
3. ALL ANCHORS AND REINFORCING BARS TO BE INSTALLED IN ACCORDANCE TO AS5216, RELEVANT ETAG CERTIFICATIONS AND MANUFACTURER'S SPECIFICATIONS.
4. ALL BOLTING ANCHORS TO BE HAS 5.8 GRADE UNO.
5. UNLESS NOTED OTHERWISE ALL CHEMICAL ANCHORS AND BOLTS TO BE CHEMICALLY ANCHORED WITH HILTI HIT RE500 OR APPROVED EQUIVALENT.
6. THOROUGHLY CLEAN THE HOLE USING A ROUND WIRE BRUSH AND BLOW OUT ALL DUST. AS PER MANUFACTURER'S SPECIFICATIONS ENSURE HOLE IS CLEAN AND DRY AND INSERT SUFFICIENT HIT RE500 EPOXY INTO THE BASE OF THE HOLE TO ENSURE THAT WHEN THE BAR IS INSTALLED EPOXY APPEARS AT THE FACE OF THE HOLE.
7. ENSURE BAR OR ANCHOR IS NOT DISTURBED WHILST EPOXY IS CURING.
8. MINIMUM FINISH: . EXTERNAL -HOT DIPPED GALVANISED . INTERNAL -ZINC PLATED

SITEWELD

1. REMOVE WELD SPATT AND SLAG.
2. CLEAN THOROUGHLY THE AREA AFFECTED BY WELDING BY HAND OR POWER SCRUBBING. ADD A VERY SMALL AMOUNT OF SODIUM CHROMATE, SODIUM DICHROMATE OR POTASSIUM DICHROMATE TO REDUCE RISK OF DEVELOPING SURFACE RUST.
3. PAINT WITH TWO COATS OF COLD-GAL OR WITH SELECTED EPOXY PAINT SYSTEM SUITABLE SYSTEM FOR THE ENVIRONMENT.

COLD-FORMED STRUCTURAL STEEL

1. ALL WORKMANSHIP AND MATERIAL SHALL BE IN ACCORDANCE WITH AS4600 EXCEPT WHERE VARIED BY THE CONTRACT DOCUMENTS
2. UNLESS NOTED OTHERWISE, ALL STEEL SHALL BE IN ACCORDANCE WITH AS1163, AS1397 (EXCLUDING GRADE G550, LESS THAN 0.9mm IN THICKNESS), AS/NZS 1594, AS/NZS 1595 AND AS/NZS 3678, AS APPROPRIATE.
3. BOLTS ARE DESIGNATED ON THE DRAWINGS BY THE NUMBER, DIAMETER, GRADE AND TIGHTENING PROCEDURE. STEEL BOLTS, NUTS AND WASHERS SHALL COMPLY WITH AS1110.1, AS1111.1, AS1112.1, AS1112.2, AS1112.3, AS1112.4, AS/NZS 1252, AS/NZS 1559 AND AS 4291.1 (ISO 898-1), AS APPROPRIATE.
4. FILLET WELDS SHALL BE 3mm CONTINUOUS USING ELECTRODES AS1554.1 U.N.O BUTT WELDS SHALL BE COMPLETE PENETRATION BUTT WELDS IN ACCORDANCE WITH AS1554.1. ALL WELDS AND WELD TESTING, INCLUDING EXTEND OF NON-DESTRUCTIVE EXAMINATION AND REPAIR OF DEFECTIVE WELDS SHALL BE IN ACCORDANCE WITH AS1554.1. PURLINS & GIRTS
5. COLD ROLLED PURLINS/GIRTS ARE TO BE INSTALLED COMPLETE WITH BRIDGING/GIRT FEET/ETC AND BOLTS WITH INTEGRAL WASHERS IN ACCORDANCE WITH THE MANUFACTURER'S SPECIFICATIONS AND RECOMMENDATIONS.
6. UNLESS NOTED OTHER WISE, ALL BOLTS SHALL BE M12 CATEGORY 4.6/S. UNLESS NOTED OTHERWISE, NO CONNECTION SHALL HAVE LESS THAN 2 BOLTS. ALL BOLTS AND WASHERS SHALL BE GALVANISED. UNLESS NOTED OTHERWISE, CLEATS AND GUSSETS SHALL BE 6mm THICK.
7. COLD FORMED STEELWORK ABOVE GROUND SHALL HAVE THE FOLLOWING SURFACE TREATMENT: AS2312.2-2014 SYSTEM: Z350 HOT DIP ZINC COATED SHEET IN ACCORDANCE WITH AS1397-2011.
8. THE SUSPENSION OF CEILINGS, SERVICES ETC FROM PURLINS SHALL BE IN ACCORDANCE WITH THE APPROVED METHODS DETAILED BELOW;



SWIMMING POOL

1. POOL BASE TO BE FOUNDED INTO VIRGIN MATEFIAL. SUPPORTING FOUNDATION MATEFIAL TO HAVE UNIFORM MOISTURE CONTENT.
2. THIS DESIGN IS SUITABLE FOR CLASS 'A', 'S' & 'M' SITES CLASSIFIED UNDER AS2870-2011 RESIDENTIAL SLABS AND FOOTINGS WITH A MINIMUM SAFE BEARING CAPACITY OF 300kPa@ UNDERSIDE POOL.
3. PROVIDE AN EXPANSION JOINT BETWEEN THE SWIMMING POOL STRUCTURE AND ANY OTHER STRUCTURE SUCH AS CONCRETE WALKWAYS, FOOTING, ETC.
4. ALL VERTICAL BARS, 70min Cover (MUST BE WATER SIDE) UNLESS OTHERWISE NOTED.
5. ALL HORIZONTAL BARS MUST BE TIED TO BACK OF VERTICAL BARS, UNLESS OTHERWISE NOTED.
6. VERTICAL BARS COGGED INTO COPING AND WALKWAYS, 50mm COVER (TOP). CROSS BARS TIED TO UNDERSIDE.
7. CURING: THE POOL MUST BE COVERED WITH FULLY TAPED, CLEAR POLYTHENE SHEETING, WITH ALL EDGES FULLY HELD DOWN TO PREVENT WIND GETTING UNDER AND DRYING OUT, FOR A MINIMUM OF 14 DAYS.
8. IF ROCK IS FOUND, THE WHOLE POOL IS TO BE PILED TO ROCK. REFER TO THE ENGINEER FOR INSTRUCTIONS.
9. SEDIMENT CONTROLS TO BE PROVIDED AROUND ALL EXCAVATION WORKS. REFER TO LOCAL COUNCIL REQUIREMENTS.
10. WHERE IT IS CONSIDERED THAT GROUND WATER CAN BUILT UP TO A LEVEL 500mm ABOVE THE FLOOR OF THE EXCAVATION ADEQUATE DRAINAGE SHALL BE PROVIDED UNDER THE POOL FLOOR.
11. CONCRETE TO HAVE A MINIMUM DESIGN STRENGTH OF fc=32MPa AT 28 DAYS, CONCRETE TO BE PNEUMATICALLY APPLIED.
12. UPON COMPLETION OF CONCRETE THE HYDROSTATIC VALVE IS TO BE CHECKED TO ENSURE EFFECTIVE AND SUFFICIENT OPERATION.
13. ALL REINFORCEMENT TO BE FIRMLY SUPPORTED BY BAR CHAIRS.
14. THE DESIGN ENGINEER MUST BE CONTACTED TO INSPECT & APPROVE ALL STRUCTURAL ELEMENTS,(INCLUDING SLAB REINFORCEMENT) SHOWN ON THIS DRAWING, BEFORE BEING COVERED BY THE NEXT STAGE CONSTRUCTION.
15. CERTIFICATION IS DEPENDENT ON THE ABOVE CONDITIONS.
16. SWIMMING POOL CONCRETE GRADE TO BE 32MPa AND PNEUMATICALLY APPLIED.

REV.	DATE	AMENDMENT DESCRIPTION	DRAWN
B	08.09.2026	FOR APPROVAL	TA
A	28.03.2026	FOR APPROVAL	TA

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PROPOSED DUAL-OCCUPANCY
18 CARRAMAR CRESCENT, MIRANDA NSW

NOTES-1

DRAWN	CHECKED	SCALE	DATUM	DRAWING No.	REV.
TA	TONY AHAL - Civil & Structural Engineer B.E. Civil (USyd) (MIEAust):1080225	AS SHOWN	N/A	S018C - S1A	B

MASONRY

1. ALL WORKMANSHIP AND MATERIALS SHALL BE IN ACCORDANCE WITH AS 3700.
2. STRENGTH AND DURABILITY REQUIREMENTS FOR MASONRY AND EMBEDDED ITEMS ARE SPECIFIED IN THE PROJECT INFORMATION TABLE.
3. 3. ALL BRICKS TO HAVE A MINIMUM CHARACTERISTIC COMPRESSIVE STRENGTH OF 20 MPa
4. ALL BLOCKS TO HAVE A MINIMUM CHARACTERISTIC COMPRESSIVE STRENGTH OF 15 MPa.
5. ONLY LOAD BEARING MASONRY WALLS ARE SHOWN UNDER CONCRETE SLABS.
6. MASONRY SUPPORTING SLABS AND BEAMS SHALL BE TROWELED SMOOTH WITH MORTAR FILLING ALL VOIDS. TWO LAYERS OF RIGID DPC SHALL BE PLACED FULL WIDTH ACROSS SUCH LOAD BEARING SURFACES EXCEPT WHERE PROPRIETARY BEARING STRIPS IS NOTED OR ALTERNATIVE DETAIL IS DOCUMENTED.
7. THE HEADS OF LOAD BEARING WALLS SHALL NOT EXTEND ABOVE THE SOFFIT OF THE CONCRETE SLAB ABOVE.
8. 8. ALL DOUBLE SKIN SOLID WALLS SUCH AS 230mm THICK BRICKWORK SHALL BE BONDED BY A HEADER COURSE EVERY 4th COURSE.
9. ALL MASONRY SUPPORTING OR SUPPORTED BY CONCRETE FLOORS SHALL BE PROVIDED WITH VERTICAL JOINTS TO MATCH ANY CONTROL JOINTS IN THE CONCRETE.
10. NON LOAD BEARING WALLS SHALL BE SEPARATED FROM CONCRETE ABOVE BY 20mm THICK CLOSED CELL POLYSTYRENE STRIP, FIRE RATED AS REQUIRED.
11. NO CHASES OR RECESSED ARE PERMITTED IN STRUCTURAL MASONRY WITHOUT THE APPROVAL OF THE ENGINEER.
12. IT IS THE CONTRACTORS RESPONSIBILITY TO CONFIRM MASONRY JOINT LOCATIONS WITH THE BRICKLAYER AND ARCHITECT. REFER TO ARCHITECTS DOCUMENTATION FOR THE LOCATION OF ALL MASONRY JOINTS AS A MINIMUM PROVIDE THE ARTICULATION, CONTRACTION OR EXPANSION JOINTS AS SPECIFIED IN 4773.1. OR AS3700 AS APPROPRIATE
13. REFER TO CONCRETE NOTES FOR DE-PROPPING PRIOR TO CONSTRUCTION OF MASONRY WALLS ON SUSPENDED SLABS.
14. BRICK TIES SHALL BE SPACED NOT GREATER THAN 600mm IN EACH DIRECTION ADJACENT TO ANY SUPPORT, CONTROL JOINT OR OPENING, PROVIDE THE FIRST ROW WITHIN 300mm.
15. WHERE NON-METALLIC TIES ARE USED THE BUILDER SHALL ENSURE THAT MORTAR IS PROVIDED TOP TO BOTTOM BEFORE PLACING BRICKS. BRICK TIES SHALL COMPLY WITH AS 2699. HEAD RESTRAINT TIES TO BE PROVIDED IN ALL WALLS AT 600c/c UNO.
16. AT CONCRETE / STEEL COLUMN JUNCTIONS MET 5-3 EXPANSION TIES TO BE USED (INTERNAL SKIN OF CAVITY WALL OR INTERNAL WALL).
17. ALL RETAINING WALLS TO BE WATERPROOFED TO ARCHITECTS DETAILS PROVIDED (2 COATS OF BITUMEN PAINT AS A MINIMUM). WATERPROOFING TO BE CARRIED OUT AS SPECIFIED BY THE MANUFACTURER.
18. PROVIDE BOND BEAMS IN BLOCKWORK WALLS ABOVE ALL OPENINGS.
19. REINFORCED CONCRETE BLOCKWORK SHALL COMPLY WITH THE FOLLOWING, UNLESS NOTED OTHERWISE:
20. PROVIDE CLEANOUT HOLES 100mm SQUARE MINIMUM AT BASE OF ALL WALLS AND ROD CORE HOLES TO REMOVE PROTRUDING MORTAR FINS PRIOR TO GROUTING.
21. REINFORCEMENT PROJECTING FROM FOUNDATION OR SLABS INTO CORES, SHALL BE SET ACCURATELY IN PLACE USING TEMPLATES TO ALIGN WITH THE CENTRE OF THE LENGTH OF CORES AND WITH COVER AS NOTED. WHERE HORIZONTAL BARS ARE INDICATED, THE WEBS OF THE BLOCKS BELOW THE BARS SHALL BE CUT DOWN TO ACCOMMODATE THE BARS.
22. GROUT ALL CORES IN REINFORCED BLOCKWORK UNLESS OTHERWISE NOTED. HEIGHT OF BLOCKWORK TO BE GROUTED ON ONE DAY SHALL BE 2400mm. GROUT SHALL BE PLACED IN LIFTS OF 1200mm MAXIMUM AND COMPACTED BY POKER VIBRATOR. ALLOW 4 HOURS BETWEEN SUCCESSIVE LIFTS TO ALLOW PLASTIC SETTLEMENT TO OCCUR.
23. PROVIDE 50mm COVER FROM THE OUTSIDE OF THE BLOCKWORK U.N.O.
24. QDERECOMMENDS THE USE OF PROPRIETY PLASTIC CHAIRS FOR THE ACCURATE PLACEMENT OF VERTICAL REINFORCEMENT.

COMPOSITE METAL DECKING

1. PANELS TO BE INSTALLED IN ACCORDANCE WITH MANUFACTURERS DETAILS.
2. BEFORE PANELS ARE LAID, TOP COURSE OF BRICKWORK IS TO BE STRAIGHT AND LEVEL. IF REQUIRED PROVIDE LAYER OF SMOOTH HARD MORTAR. SLIP JOINTS SHALL BE PROVIDED AT ALL MASONRY SUPPORTS UNLESS NOTED OTHERWISE.
3. BEFORE CONCRETE IS PLACED, ANY ACCUMULATED DEBRIS, GREASE OR ANY OTHER SUBSTANCE WILL NEED TO BE REMOVED TO ENSURE CLEAN BONDING SURFACE. ANY PONDED RAINWATER SHOULD BE REMOVED.
4. UNLESS NOTED OTHERWISE PROPPING OF METAL DECKING SHALL BE IN ACCORDANCE WITH MANUFACTURERS
5. PROPS SHOULD NOT BE REMOVED UNTIL CONCRETE HAS REACHED SPECIFIED DESIGN STRENGTH

CONTRACTOR DESIGNED PILES

1. THE PILES ARE TO BE DESIGNED, INSTALLED AND CERTIFIED BY THE 1. CONTRACTOR FOR THE LOADS INDICATED ON THE DRAWINGS. THE DESIGN LIFE OF THE PILES IS TO BE A MINIMUM OF 50 YEARS
2. UNLESS NOTED OTHERWISE IN THE PROJECT INFORMATION TABLE.
3. DESIGN AND INSTALLATION OF PILES IS TO BE IN ACCORDANCE WITH AS2159 AND TO INCLUDE ALLOWANCE FOR INDUCED MOMENTS IN PILES DUE TO CONSTRUCTION TOLERANCES.

CONCRETE PIERS

1. PIER DIAMETER AND LOCATIONS ARE SHOWN ON PLAN.
2. U.N.O MINIMUM PIER DEPTH IS 600 MM BELOW FOOTINGS AND WHENEVER NOMINATED SHOULD BE SOCKETED A MINIMUM 300 MM INTO STIFF CLAY.
3. ALL PIER HOLES SHALL BE CLEANED AND DEWATERED PRIOR THE PLACEMENT OF CONCRETE.
4. ALL PIERS SHALL BE POURED SEPERATE TO FOOTINGS.
5. IT SHOULD BE NOTED IF ANY OF THE FOOTINGS ENCOUNTER ROCK OR SHALE, THEN ALL FOOTINGS SHALL BE PIERED ROCK OR SHALE. IF PARTIALLY PIERED TO ROCK THEN ARTICULATION JOINTS SHALL BE PROVIDED AT THE ROCK/NON-ROCK INTERFACE.
6. CONCRETE PIERS/FOOTINGS ARE REQUIRED TO UNIFORM BEARING. ENGINEER OR CERTIFIER SHALL INSPECT GROUND BEARING OF PIERS/FOOTINGS PRIOR TO POURING OF CONCRETE.

RESIDENTIAL SLABS AND FOOTINGS

1. THIS DESIGN HAS BEEN CARRIED OUT IN ACCORDANCE WITH AS 2870. CONFIRMATION OF THE SITE CLASSIFICATION IN ACCORDANCE WITH AS 2870 IS REQUIRED BY A GEOTECHNICAL ENGINEER PRIOR TO CONSTRUCTION COMMENCING ON SITE AND SHALL INCLUDE CONFIRMATION OF BEARING CAPACITY.
2. THE FOUNDATION MATERIAL SHALL BE UNIFORM AND PIERS OR BLOCKDOWNS SHALL BE USED AT THE DIRECTION OF THE GEOTECHNICAL ENGINEER TO ACHIEVE UNIFORM BEARING WHERE REQUIRED.
3. THE OWNERS ATTENTION IS DRAWN TO THE ACCEPTABLE LEVELS OF FOUNDATION PERFORMANCE AS OUTLINED BY AS 2870. ACCORDINGLY CATEGORY 1 OR 2 DAMAGE MAY BE EXPECTED UNDER SOME CONDITIONS. SHOULD A HIGHER LEVEL OF CRACK CONTROL BE REQUIRED THEN THE ENGINEER SHOULD BE NOTIFIED SO THAT THIS CAN BE INCORPORATED INTO THE DESIGN.
4. SITES SHALL BE PREPARED IN ACCORDANCE WITH AS 2870. AS A MINIMUM THIS MUST INCLUDE: TOP SOIL CONTAINING GRASS, ROOTS OR ANY OTHER ORGANIC MATERIAL SHALL BE REMOVED FROM THE AREA ON WHICH THE SLAB IS TO REST. BLIND WITH SAND SUFFICIENT ONLY TO ENSURE NO DAMAGE TO DAMP PROOF MEMBRANE (MAXIMUM 50mm). A VAPOUR BARRIER/DAMP PROOF MEMBRANE SHALL BE PROVIDED. THE MEMBRANE SHALL BE POLYTHENE SHEETING OF MINIMUM THICKNESS 0.2mm. THE SHEET SHALL BE LAID BENEATH THE SLAB SUCH THAT THE SLAB AND ALL BEAMS ARE ENTIRELY UNDERLAID. THE SHEET SHALL EXTEND UNDER EDGE BEAMS TO GROUND LEVEL. SHEET JOINTS SHALL BE FULLY TAPED LAPS WITH A MINIMUM OF 200mm OVERLAP. PENETRATIONS AT PIPES OR FITTINGS, ETC SHALL BE TAPED OR SEALED WITH A CLOSE-FITTING SLEEVE OR TURN UP OF THE MEMBRANE.
5. WHERE NOTED ON DRAWINGS AS CONTROLLED FILL, THE FILLING SHALL COMPLY WITH THE FOLLOWING: SAND FILL UP TO 0.8M DEEP THAT IS WELL COMPACTED BY A VIBRATING PLATE OR VIBRATING ROLLER IN LAYERS NOT MORE THAN 300mm THICK. NON-SAND FILL UP TO 0.4M DEEP THAT IS WELL COMPACTED BY A MECHANICAL ROLLER IN LAYERS NOT MORE THAN 150mm THICK. ANY CLAY SHALL BE MOIST DURING COMPACTION.
6. WHERE NOTED ON DRAWINGS AS ROLLED FILL, THE FILLING SHALL BE MATERIAL COMPACTED IN LAYERS BY REPEATED ROLLING WITH AN EXCAVATOR OR SIMILAR EQUIPMENT. THE DEPTH OF THE ROLLED FILL SHALL NOT EXCEED: 0.6M FOR SAND FILL COMPACTED IN LAYERS NOT MORE THAN 300mm. & 0.3M FOR NON-SAND FILL COMPACTED IN LAYERS NOT MORE THAN 150mm.
7. WHERE NOTED ON DRAWINGS AS `FILL AS FORMWORK', FILLING SHALL BE COMPACTED SUFFICIENTLY TO PROVIDE A STABLE PLATFORM DURING CONSTRUCTION. THIS WOULD NORMALLY BE PROVIDED BY COMPACTION TO AT LEAST 85% MAXIMUM STANDARD DRY DENSITY.
8. HEATING CABLES AND HOT WATER PIPES, ELECTRIC HEATING CABLES MAY BE EMBEDDED IN THE SLAB WITHOUT ANY INCREASE IN THICKNESS OR REINFORCEMENT. WHERE HOT WATER PIPES ARE TO BE EMBEDDED IN A SLAB, THE SLAB THICKNESS SHALL BE INCREASED BY 25mm AND THE MESH SIZE INCREASED ONE LEVEL (E.G FROM SL72 TO SL82).
9. ALLOTMENTS CONTAINING REACTIVE SITES CLASSIFIED AS M, H1, H2 OR E SHALL BE PROVIDED WITH AN ADEQUATE SYSTEM OF DRAINING IN ACCORDANCE WITH AS 2870 TO ENSURE BEST POSSIBLE FOUNDATION PERFORMANCE. AT A MINIMUM THE FOLLOWING SHOULD BE MAINTAINED: THE SITE SHOULD BE GRADED OR DRAINED SO THAT WATER CANNOT POND AGAINST OR NEAR THE BUILDING. THE GROUND IMMEDIATELY ADJACENT THE BUILDING SHOULD BE GRADED TO A UNIFORM FALL OF 50mm MINIMUM AWAY FROM THE BUILDING OVER THE FIRST METRE. SUBFLOOR AREAS SHOULD BE GRADED TO PREVENT WATER PONDING WHICH MAY AFFECT FOOTING PERFORMANCE.

INSITU BORED CONCRETE PILES

1. ALL WORKMANSHIP AND MATERIALS SHALL BE IN ACCORDANCE WITH AS2159 PILING DESIGN & INSTALLATION CURRENT EDITIONS WITH AMENDMENTS, EXCEPT WHERE VARIED BY THE CONTRACT DOCUMENTS.
2. ALL DESIGN AND INSTALLATION SHALL BE COMPLETED BY AN EXPERIENCED CONTRACTOR SPECIALISING IN FOUNDATION ENGINEERING AND SHALL BE IN ACCORDANCE WITH AS2159.SUBMIT DETAILS OF PROPOSED DRILLING METHODS, EQUIPMENT AND SEQUENCE. GIVE NOTICE SO THAT INSPECTION MAY BE MADE OF THE FOLLOWING AS APPLICABLE:
   - AT COMPLETION OF EXCAVATION OF PIERS
   - REINFORCEMENT OF PIER
3. SETTING OUT - PEG THE POSITION OF EACH PILE AND ESTABLISH A GRID OF RECOVERY PEGS TO ENABLE THE SETTING OUT TO BE CHECKED AT ANY TIME. PILES AND OR PIERS SHALL BE LOCATED WITHIN 75mm OF THE PLAN LOCATIONS AS SHOWN ON THE ENGINEERING DRAWINGS.
4. PIERS SHALL BE POURED WITHIN 24 HOURS OF EXCAVATION UNLESS OTHERWISE AGREED.
5. PIER WALL SHALL BE MAINTAINED STABLE PRIOR TO POURING CONCRETE.
6. PIER BASES SHALL BE LEVEL AND FREE OF ALL LOOSE MATERIAL.
7. REMOVE ALL FREE WATER FROM PIERS BEFORE POURING.
8. CONFIRM BEARING PRESSURE AT BASE OF ALL PIERS BY GEOTECHNICAL TESTING/INSPECTIONS.
9. CONCRETE SHALL BE PLACED IN SUCH A MANNER SO AS TO AVOID SEGREGATION.
10. REFER ALSO TO CONCRETE NOTES.
11. THE PILE DEPTHS SHOWN ON THE ENGINEERS DRAWINGS ARE PROVISIONAL. ACTUAL PILE LENGTHS ARE TO BE MEASURED BY THE CONTRACTOR AND APPROVED BY THE SUPERINTENDENT.
12. OBSERVE SAFE-WORKING PRACTICES, INCLUDING THE RELEVANT PRACTICES RECOMMENDED IN AS2159 APPENDIX B.
13. PROVIDE & INSTALL FACILITIES NECESSARY FOR INSPECTION OF PILING INCLUDING SAFE ACCESS, LIGHTING, VENTILATION AND THE LIKE.
14. REFER TO DRAWINGS FOR ALL OTHER PIER INFORMATION.
15. RECORD THE RELEVANT INFORMATION AS LISTED IN AS2159. AND FORWARD TO ENGINEER/SUPERINTENDENT COPIES OF EACH RECORD TO THE SUPERINTENDENT.
16. BORED PIERS SHALL BE WITHIN 2% OF VERTICAL AND WITHIN 75 mm OF LOCATION (MEASURED IN ANY DIRECTION). THE PILING SUB-CONTRACTOR SHALL SUBMIT TO THE ENGINEER FOR REVIEW A RECTIFICATION DESIGN FOR BORED PIERS OUT OF LOCATION.

WATERPROOFING

1. REFER TO ARCHITECTS DOCUMENTATION FOR ALL CAPPING, FLASHINGS & WATERPROOFING DETAILS.
2. WHERE THE ARCHITECTURAL SPECIFICATION REQUIRES WATERPROOF CONCRETE PROVIDE A WATERPROOF CONCRETE MIX USING WATERPROOFING ADMIXTURES IN ACCORDANCE WITH THE MANUFACTURERS SPECIFICATION.

CONSTRUCTION JOINTS

1. CONSTRUCTION JOINTS SHALL BE USED ONLY WHERE SHOWN ON THE DRAWINGS, AND NO CONSTRUCTION JOINT SHOWN ON THE DRAWINGS SHALL BE OMITTED UNLESS APPROVED IN WRITING BY THE SUPERINTENDENT. LOCATION OF CONSTRUCTION JOINTS ARE SHOWN THUS CJ.
2. CONSTRUCTION JOINTS WHERE NOT SHOWN SHALL BE LOCATED TO THE APPROVAL OF THE DESIGN ENGINEER.

FRAMING AND BRACING

1. ALL BRACING AND TIE DOWNS TO BE IN ACCORDANCE WITH AS1684
2. UNLESS NOTED OTHERWISE PROVIDE THE MINIMUM FOR

2.1. ROOFS AND HORIZONTAL BRACING -USE DIAGONAL CROSS STRAP BRACING THROUGHOUT ROOF. SCREW FIX TO EACH RAFTER,PURLIN OR TRUSS AND SECURELY FASTEN TO TOP PLATE/BEAM AT ENDS.

2.2. STUD WALLS -STRAP TOP & BOTTOM PLATE TO A MINIMUM OF EVERY SECOND STUD SECURE FLOOR PLATE TO SLAB BELOW WITH M12 CHEMICAL ANCHORS AT 1200mm MAX CTS OR SECURE EVERY SECOND STUD TO FLOOR BEARERS BELOW

2.3. MASONRY WALLS - FASTEN TOP PLATE TO BRICKWORK USING ROOF STRAPS (SIMILAR M.E.T. 7.1 OR APPROVED EQUIVALENT) AT A MAXIMUM OF 1200mm SPACING ANCHORED 15 C/S DOWN PROVIDE STRAPS EACH SIDE OF ALL OPENINGS PROVIDE ADDITIONAL HOLD DOWNS AS DETAILED

2.4. VERTICAL BRACING -PROVIDE VERTICAL BRACING IN ACCORDANCE WITH AS 1684. PROVIDE ADDITIONAL VERTICAL BRACING WHERE SHOWN AND AS DETAILED ON THESE DRAWINGS

LINTELS

1. UNLESS OTHER SUPPORT IS SPECIFIED, PROVIDE PROPRIETY LINTEL (PRE-CAST OR GALINTEL) INSTALLED AND PROPPED TO MANUFACTURERS SPECIFICATIONS FOR ALL OPENINGS UNDER 3000mm
2. ALTERNATIVELY PROVIDE LINTELS IN ACCORDANCE WITH TABLE BELOW (ONE TO EACH LEAF OF WALL)
3. REFER TO PROJECT DESIGN INFORMATION TABLE FOR DURABILITY REQUIREMENT.

BRICK LINTEL SCHEDULE			
OPENING SIZE (mm)	INTERNAL SKIN	EXTERNAL SKIN	END BEARING
UP TO 900mm	100 x 8mm FLAT BAR	100 x 6mm FLAT BAR	100 mm
1200	100 x 10mm FLAT BAR	100 x 8mm FLAT BAR	100 mm
1500	100 x 100 x 8mm ANGLE	100 x 100 x 6mm ANGLE	150 mm
2100	150 x 100 x 8mm ANGLE	150 x 100 x 6mm ANGLE	150 mm
2400	150 x 100 x 8mm ANGLE	150 x 100 x 8mm ANGLE	150 mm
2700	150 x 100 x 10mm ANGLE	150 x 100 x 10mm ANGLE	150 mm
3000	150 x 100 x 12mm ANGLE	150 x 100 x 12mm ANGLE	150 mm

NOTES
- ALL STEEL LINTELS TO BE HOT DIPPED GALVANIZED.
- ABOVE ALL LINTELS, PROVIDE M.E.T MASONARY REINFORCEMENT FOR 2 COURSES AND CONTINUE 1000mm PAST OPENING.

STEELWORK

1. ALL STEELWORK WORKMANSHIP, MATERIALS, FABRICATION & INSTALLATION SHALL BE IN ACCORDANCE WITH AS4100 STEEL STRUCTURES CODE, AS1554, AS1163, AS3679 AND AS4600 EXCEPT WHERE VARIED BY THE CONTRACT DOCUMENTS.
2. STEEL QUALITY SHALL CONFORM WITH THE FOLLOWING AUSTRALIAN STANDARDS UNO: · STEEL PLATES & SECTIONS AS3679 GRADE 300 · RECTANGULAR HOLLOW SECTIONS AS1163 GRADE C350 LO · CIRCULAR HOLLOW SECTIONS AS1163 GRADE C350 · SQUARE HOLLOW SECTIONS AS1163 GRADE C350 · PURLINS/GIRTS AS1397 GRADE G450 Z350
3. ALL WELDING SHALL BE IN ACCORDANCE WITH THE PROVISIONS OF AS1554.
4. ALL WELDS SHALL BE 6mm CONTINUOUS FILLET TYPE SP CATEGORY MADE WITH E48XX ELECTRODE OR W50X OR EQUIVALENT UNO.
5. FOR STEEL LESS THAN 6mm THICK, FILLET WELD THICKNESS TO MATCH THE BASE METAL THICKNESS UNO.
6. ALL BUTT WELDS SHALL BE FULL PENETRATION TYPE SP CATEGORY UNO.
7. WELD INSPECTION IS REQUIRED IN ACCORDANCE WITH AS/NZS 1554.1. · ALL GP / SP WELDS SHALL BE 100% VISUALLY SCANNED UNO. · SP FILLET WELDS SHALL HAVE 10% VISUAL EXAMINATION UNO. · SP BUTT WELDS SHALL HAVE 50% VISUAL EXAMINATION UNO. · ALL GP WELDS SHALL HAVE 10% VISUAL EXAMINATION UNO.
8. ALL CLEAT, GUSSET, FIN, END PLATES etc. SHALL BE 10mm THICK UNO.
9. BUTT WELD ALL FLANGES AT END PLATES AND AT ALL MITRE JOINTS.
10. ALL MEMBERS SHALL BE SUPPLIED IN SINGLE LENGTHS. SPLICES SHALL ONLY BE PERMITTED IN LOCATIONS SHOWN ON THESE STRUCTURAL DRAWINGS.
11. PROVIDE SEAL PLATES TO ALL HOLLOW SECTIONS.
12. NO PENETRATIONS OR NOTCHES OTHER THAN THOSE SHOWN ON THE STRUCTURAL DRAWINGS SHALL BE MADE IN ANY STEEL MEMBERS WITHOUT APPROVAL OF THE STRUCTURAL ENGINEER.
13. BOLT DESIGNATION: 4.6/S: COMMERCIAL BOLTS OF GRADE 4.6 TO AS 1110, SNUG TIGHTENED. 8.8/S: HIGH STRENGTH STRUCTURAL BOLTS OF GRADE 8.8 TO AS 1252, SNUG TIGHTENED. 8.8/TB: HIGH STRENGTH STRUCTURAL BOLTS OF GRADE 8.8 TO AS 1252, FULLY TENSIONED TO AS 4100 AS A BEARING JOINT. 8.8/TF: HIGH STRENGTH STRUCTURAL BOLTS OF GRADE 8.8 TO AS 1252, FULLY TENSIONED AS A FRICTION JOINT WITH BEARING SURFACES LEFT UNCOATED.
14. ALL BOLTS SHALL BE M20 GRADE 8.8/S UNO.
15. NO CONNECTION SHALL HAVE LESS THAN 2 BOLTS UNO.
16. ALL BOLTS, NUTS AND WASHERS SHALL BE HOT DIP GALVANISED.
17. BOLTS GRADED 4.6 OR 8.8 ARE NOT SUPPLIED AS WELDABLE ITEMS IN THEIR RESPECTIVE PRODUCT STANDARDS AND SHOULD NOT BE WELDED.
18. ENSURE ALL BOLTS HAVE SUFFICIENT LENGTH SO THAT AT LEAST ONE FULL THREAD IS EXPOSED BEYOND THE NUT AFTER TIGHTENING.
19. INTERNAL STEELWORK SHALL BE BLAST CLEANED TO CLASS 1 AND COATED WITH MINIMUM 100µm HIGH BUILD ALKYD PRIMER BEFORE ERECTION.
20. EXTERNAL STEELWORK SHALL BE BLAST CLEANED TO CLASS 2.5 AND HOT DIPPED GALVANISED TO AS4680 WITH A COATING MASS OF 600g/m² (AVERAGE).
21. AS AN ALTERNATIVE TO GALVANISING, STRUCTURAL STEELWORK MAY BE GRIT BLASTED TO CLASS 2.5 AND PAINTED WITH PROTECTIVE COATING (WITH A DURABILITY OF MIN. 25+ YEARS TO FIRST MAINTENANCE) TO MANUFACTURER'S SPECIFICATIONS TO SUIT APPROPRIATE ATMOSPHERIC CORROSIVITY CATEGORIES BASED ON AS/NZS 2312.1:2014
22. ALL STRUCTURAL STEEL MEMBERS SPECIFIED ON THE DRAWINGS OR OTHER RELATED CONTRACTS AS BEING GALVANISED SHALL CONFORM TO THE STANDARDS OF AS4680 & NCC VOLUME 2 HOUSING PROVISIONS STANDARD 2022 CI. 6.3.9
23. PROVIDE PROTECTIVE COATING ON ALL STEELWORK PARTIALLY EMBEDDED IN CONCRETE UNO. THE COATING SHOULD BE NON-CONDUCTIVE AND EXTEND A MINIMUM OF 200mm INTO THE CONCRETE AND A MINIMUM OF 100mm ABOVE THE CONCRETE IN ACCORDANCE WITH AS2312.
24. UNLESS NOTED OTHERWISE, PROTECTIVE COATINGS FOR BUILT-IN COMPONENTS FOR MASONRY CONSTRUCTION, INCLUDING (BUT NOT LIMITED TO) WALL TIES, MASONRY ANCHORS, CONNECTORS, SHELF ANGLES, LINTEL BARS, BED JOINT MESH, BOLTS & FIXINGS SHALL BE ACCORDING TO AS3700, AS2312,AS2699, & NCC VOLUME 2 HOUSING PROVISIONS STANDARD 2022 CI. 5.6.7b.
25. THE BUILDER SHALL PROVIDE ALL CLEATS AND DRILL HOLES NECESSARY FOR FIXING STEEL TO STEEL AND TIMBER TO STEEL WHETHER OR NOT DETAILED ON THE DRAWINGS.
26. PROVIDE TEMPORARY BRACING TO MAINTAIN STABILITY OF STEELWORK DURING CONSTRUCTION.
27. CHIP ALL WELDS FREE OF SLAG.
28. DO NOT GROUT UNDER BASE PLATES UNTIL FIRST LEVEL STEELWORK IS PLUMB AND FIXED BY WELDING OR BOLTING.

STEELWORK FINISHES

LOCATION	TYPE	CODE
INTERNAL	PAINT SYSTEM	ALK2
BUILT INTO MASONRY	GALVANISED	HDG 600
EXTERNAL	GALVANISED	HDG 600

SPECIFICATIONS
- ALK2 = 75 MICRONS OF EPOXY ZINC PHOSPHATE .
- ALL GALVANISING OF STRUCTURAL STEELWORK SHALL BE PROCESSED IN ACCORDANCE WITH AS4680/2006.

STRUCTURAL STEEL MEMBERS THAT ARE BUILT INTO MASONRY WALL (AS2699.3 - 2020)

LOCATION	DURABILITY CLASS	TYPICAL DISTANCE	
		SURF COAST	SHELTERED COAST
EXTERNAL & INTERNAL	R1 & R2	>10Km	>1Km
	R3	1Km TO 10Km	100m TO 1Km
	R4	<1Km	<100m

STRUCTURAL STEEL MEMBERS THAT ARE BUILT INTO MASONRY WALL (AS2699.3 - 2020)

DURABILITY CLASS OF LINTEL IN ACCORDANCE WITH AS 2699.3	MATERIAL OR PROTECTIVE REQUIREMENTS IN ACCORDANCE WITH AS 2699.3
R1,R2	HOT DIP GALVANISED WITH A MINIMUM AVERAGE COATING THICKNESS OF 300 g/m²; OR STAINLESS STEEL 316L
R3	HOT DIP GALVANISED WITH A MINIMUM AVERAGE COATING THICKNESS OF 600 g/m²; OR STAINLESS STEEL 316L
R4	STANLESS STEEL 316L OR DUPLEX COATING (HOT DIP GALVANIZED PLUS PAINT COATING) CONSISTING OF AS/NZS 4680 HDG 600 (MINIMUM AVERAGE COATING THICKNESS = 85 µm) FOLLOWED BY SWEEP BLAST CLEANING & A PAINT COATING CONSISTING OF EITHER: a) 350 µm OF HIGH BUILD EPOXY PAINT TO AS/NZS 3750.14; OR b) 250 µm OF HIGH BUILD EPOXY PAINT TO AS/NZS 3750.14 FOLLOWED BY 100 µm OF POLYURETHANE TO AS/NZS 3750.6 OR 100 µm OF ACRYLIC GLOSS TO AS/NZS 3750.5
TABLE NOTES: 1) ADDITION DECORATIVE COATINGS CAN BE APPLIED, BUT MUST NOT BE CONSIDERED FOR THE PURPOSE OF SATISFYING THE REQUIREMENTS OF THIS TABLE. 2) ANY LINTEL WITH A COATING THAT IS MODIFIED, I.E. BY CUTTING, WELDING, OR WHERE DAMAGED, MUST HAVE THE COATING RESTORED TO PROVIDE AN EQUIVALENT LEVEL OF PROTECTION PROVIDED BY THE ORIGINAL COATING.	

STRUCTURAL STEEL MEMBERS THAT ARE NOT BUILT INTO MASONRY WALL (AS4312-2008)

LOCATION	ENVIRONMENTAL EXPOSURE CLASS	TYPICAL DISTANCE	
		SURF COAST	SHELTERED COAST
EXTERNAL	C2	>50Km	>1Km
	C3	1Km - 50Km	50m - 1Km
	C4	200m TO 1Km	<50m
	C5	<200m	N/A
INTERNAL	C1	ALL CLASS 1 BUILDING INTERNAL BEAM SHALL BE CLASSIFIED AS C1 U.N.O. AS PER AS 4100-2020 (REFER TO AS4212 TABLE 2.1 FOR OTHER BUILDING CLASSES).	

STRUCTURAL STEEL MEMBERS THAT ARE NOT BUILT INTO MASONRY WALL (AS4312-2008)

ENVIRONMENT	MINIMUM PROTECTIVE COATING		
	OPTION 1 (HOT DIP GALVANISING)	OPTION 2 (DUPLEX SYSTEM). SEE ABCB HOUSING PROVISION STANDARD-2022 TABLE 6.3.9c	OPTION 3 (PAINT). SEE TABLE ABCB HOUSING PROVISION STANDARD-2022 6.3.9b
VERY LOW (MILD STEEL CORROSION RATE<1.3g µm² YEAR - C1	HDG75	-	SURFACE PREPARATIONS & PAINTING AS PER AS2312.1, TABLE 6.1., DURABILITY: MIN. 25+YEARS TO FIRST MAINTENANCE
LOW (MILD STEEL CORROSION RATE 1.3 TO 25 µm² YEAR - C2	HDG75	-	ACL2, ACC2, IZS1, PUR2A
MEDIUM (MILD STEEL CORROSION RATE 25 TO 50 µm² YEAR) - C3	HDG225	-	ACL3, ACC4, ACC5, IZS1, PUR3, PUR4
HIGH (MILD STEET CORROSION RATE 50 TO 80 µm²YEAR) - C4	HDG450	HDG150 (5 YEARS) 4D (10-15 YEARS) OR HDG300 (10 YEARS) 2D (5-10 YEARS)	ACC6, IZS3, PUR5
VERY HIGH (MILD STEEL CORROSION RATE 80 TO 200 µm² YEAR) - C5	HDG900	HDG300 (5 YEARS) 5D (10-15 YEARS) OR HDG600 (10 YEARS) 4D (5-10 YEARS)	ACC6 (C5-M ONLY), PUR5
TABLE NOTES: HOT DIP GALVANISING & DUPLEX SYSTEMS MUST BE IN ACCORDANCE WITH AS 2312.2. PAINT SYSTEMS MUST BE IN ACCORDANCE WITH AS 2312.1.			

REV.	DATE	AMENDMENT DESCRIPTION	DRAWN
B	08.09.2026	FOR APPROVAL	TA
A	28.03.2026	FOR APPROVAL	TA

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PROPOSED DUAL-OCCUPANCY

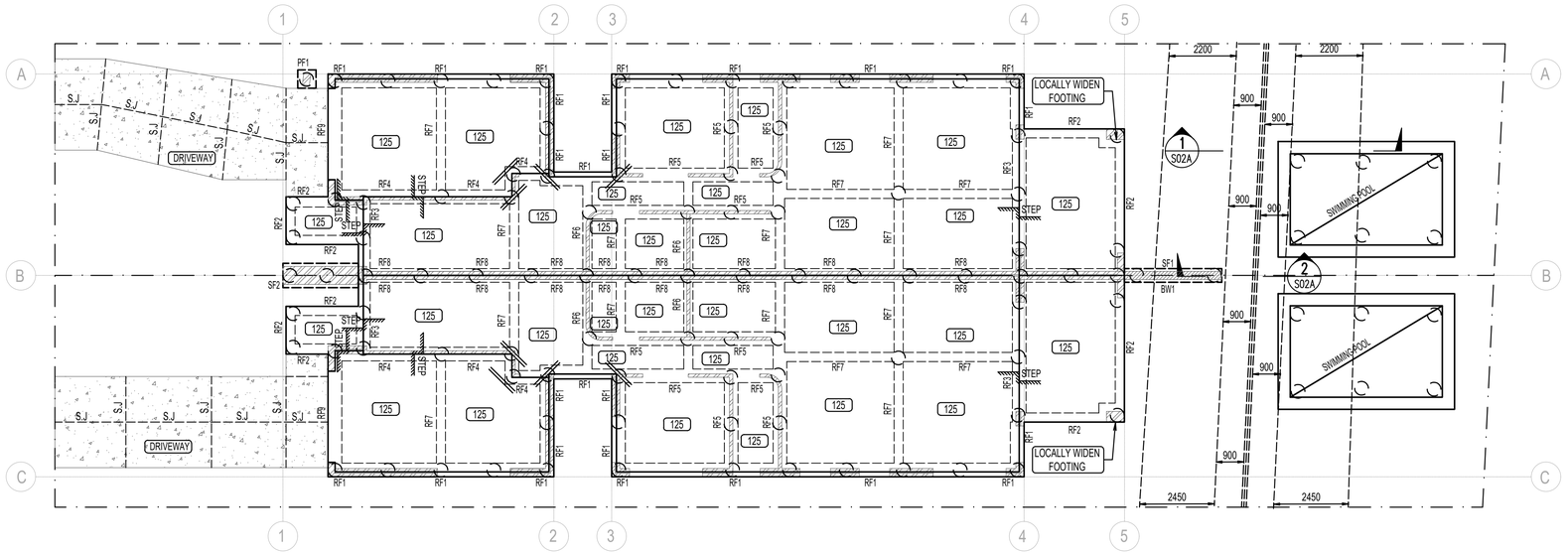
18 CARRAMAR CRESCENT, MIRANDA NSW

NOTES-2

DRAWN	CHECKED	SCALE	DATUM	DRAWING No.	REV.
TA	TONY AHAL - Civil & Structural Engineer B.E. Civil (USyd) (MIEAust) 1080225	AS SHOWN	N/A	S018C - S1B	B

GROUND FLOOR PLAN - 125/150 mm SLAB U.N.O

SCALE: 1:100



NOTES:

1. SITE IN ACCORDANCE WITH AS2870 CLASS A - ESWN-PR-2025-3559
2. SAFE BEARING CAPACITY TO BE A MN OF 150 kPa TO BE BEARING IN DENSE SAND.
3. PROVIDE SL82 MESH TOP THROUGHOUT FOR ALL 125mm THICK SLAB ON GROUND U.N.O.
4. PROVIDE SL82 MESH TOP & BOTTOM THROUGHOUT FOR ALL 150mm THICK SLAB.
5. ALL EXPOSED SLABS TO BE WATERPROOFED WITH APPROVED SYSTEMS.
6. ALL BRICKS WALLS TO BE ARTICULATED AND SPACED IN ACCORDANCE WITH AS3700 AND ARTICULATED WALLING GUIDELINE TN61
7. DRAWING TO BE READ IN CONJUNCTION WITH ARCHITECTURALS.
8. REFER TO ARCHITECTURAL DRAWINGS FOR ALL SETOUT, LEVELS, FALLS ETC.
9. REFER TO STORMWATER DRAINAGE DRAWINGS FOR LOCATION OF PUMP WELL AND SUMP PITS.
10. GEOTECHNICAL ENGINEER TO VERIFY THE SAFE BEARING CAPACITY OF FOUNDATION MATERIAL PRIOR TO CONSTRUCTION.

CONCRETE QUALITY				
ELEMENT	SLUMP	AGGREGATE (MAX. SIZE)	CEMENT TYPE	fc
SLAB ON GROUND	80mm	20mm	A	32 MPa

REINFORCEMENT COVER SCHEDULE				
MEMBER	COVER (mm)			EXPOSURE CLASSIFICATION
	TOP	BOTTOM	SIDES	
INTERNAL SLAB	30mm	30mm	30mm	A1
EXTERNAL SLAB	40mm	40mm	40mm	A2

DRIVEWAY - 120mm SLAB WITH SL82 MESH TOP U.N.O
NOTE :1. MAX 6m SAWN CONTROL JOINT
2. MAX 15m KEYED CONSTRUCTION JOINT
3. MAX 30m EXPANSION JOINT

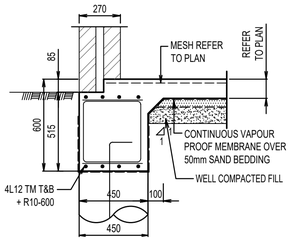
PATHWAY- 100mm SLAB WITH SL72 MESH TOP U.N.O
NOTE :1. MAX 4m SAWN CONTROL JOINT
2. MAX 10m KEYED CONSTRUCTION JOINT
3. MAX 20m EXPANSION JOINT

LEGEND	
	LOAD BEARING BRICK WALL ABOVE
	4500 BORED PIERS INTO SHALE 300MM MINIMUM WITH 1N12 EMBEDDED 300MM INTO FOOTING
125	DENOTES SLAB THICKNESS
S.J	DENOTES SAWN JOINT
150	DENOTES STEP IN SLAB
TR	2N12 1200 LONG TRIMMER BARS
BW1	DENOTES 190mm BLOCK WALL

ALL SLAB ZONES DENOTED AS 150 THICK OR 500mm ABOVE NATURAL GROUND TO HAVE SL82 MESH TOP AND BOTTOM

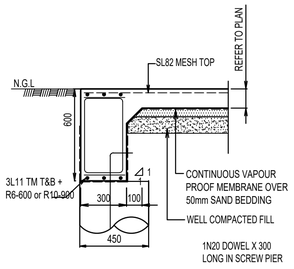
GROUND FLOOR WALLS TO BE DOUBLE BRICK CONSTRUCTION. ENGINEER TO BE NOTIFIED IF OTHERWISE

REFER TO ARCHITECTURAL DRAWINGS FOR ALL BUILDING DIMENSION. STRUCTURAL DRAWINGS ARE NOT TO BE SCALED OR USED FOR BUILDING DIMENSIONS.



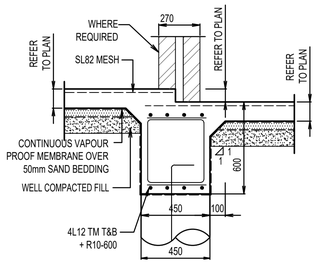
RAFT FOOTING RF1

SCALE 1:20



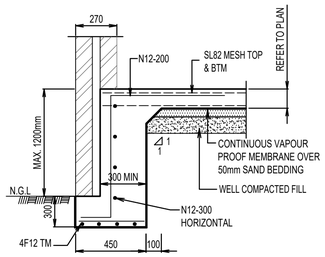
RAFT FOOTING RF2

SCALE 1:20



RAFT FOOTING RF3

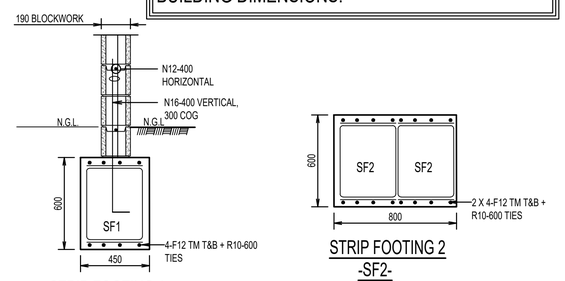
SCALE 1:20



DROP EDGE BEAM DEB1

WHERE REQUIRED

SCALE 1:20



STRIP FOOTING 1
-SF1-

STRIP FOOTING 2
-SF2-

RAFT FOOTING RF4

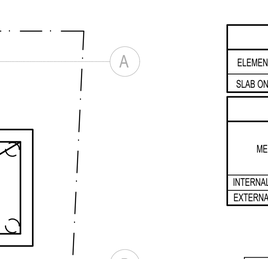
SCALE 1:20

RAFT FOOTING RF5

SCALE 1:20

RAFT FOOTING RF6

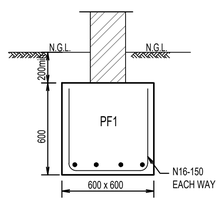
SCALE 1:20



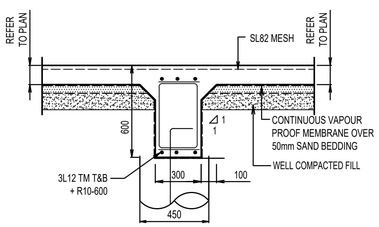
DROP EDGE BEAM DEB2

WHERE REQUIRED

SCALE 1:20

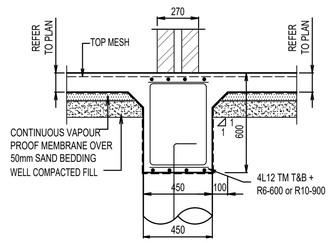


PAD FOOTING 1
-PF1-



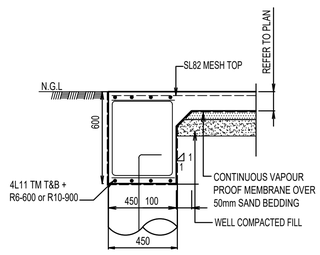
RAFT FOOTING RF7

SCALE 1:20



RAFT FOOTING RF8

SCALE 1:20



RAFT FOOTING RF9

SCALE 1:20

REV.	DATE	AMENDMENT DESCRIPTION	DRAWN
B	08.09.2026	FOR APPROVAL	TA
A	28.03.2026	FOR APPROVAL	TA

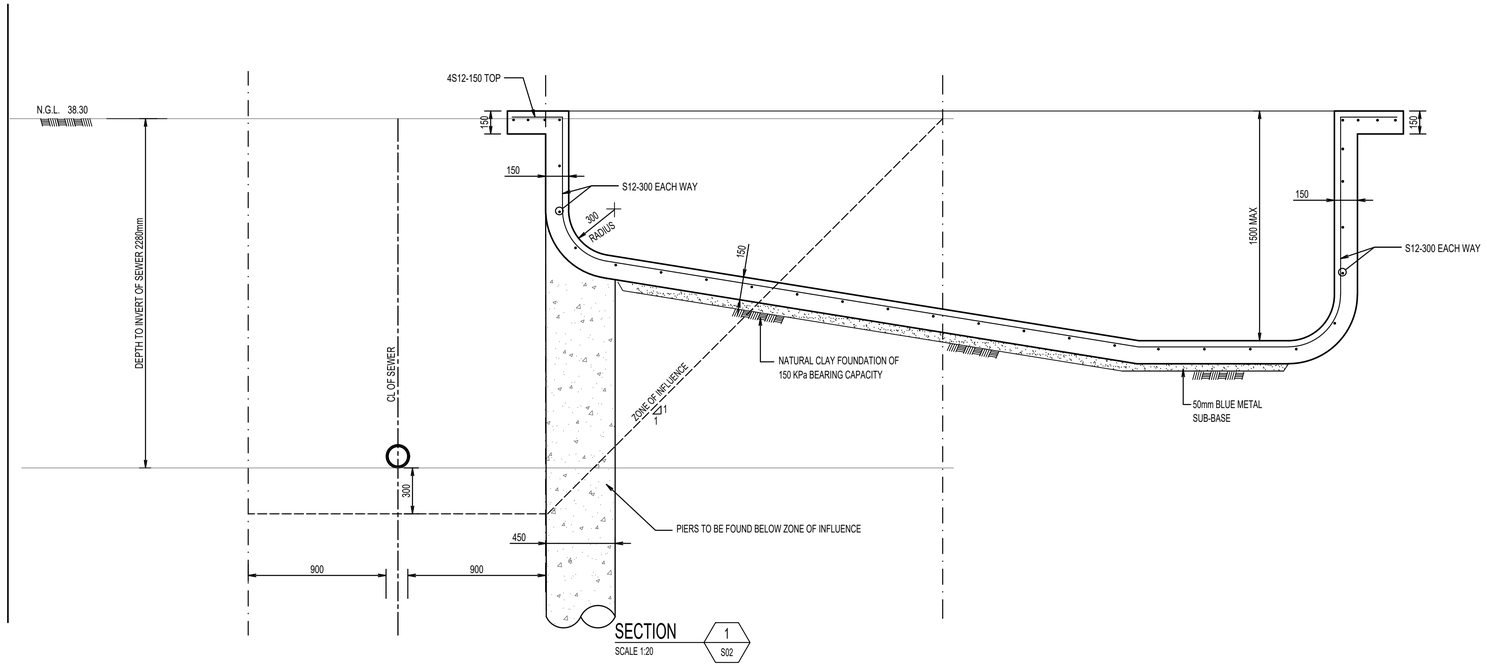
TAA CONSULTING GROUP
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PROPOSED DUAL-OCCUPANCY

18 CARRAMAR CRESCENT, MIRANDA NSW

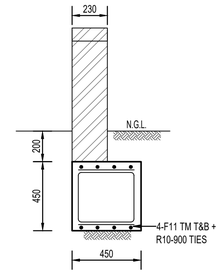
GROUND FLOOR SLAB PLAN

DRAWN	CHECKED	SCALE	DATUM	DRAWING No.	REV.
TA	TONY AHAL - Civil & Structural Engineer B.E. Civil (UtSyd) (MIEAust) 1080225	AS SHOWN	N/A	S018C - S02	B



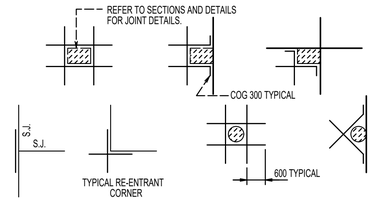
SECTION 1 / S02

SCALE 1:20



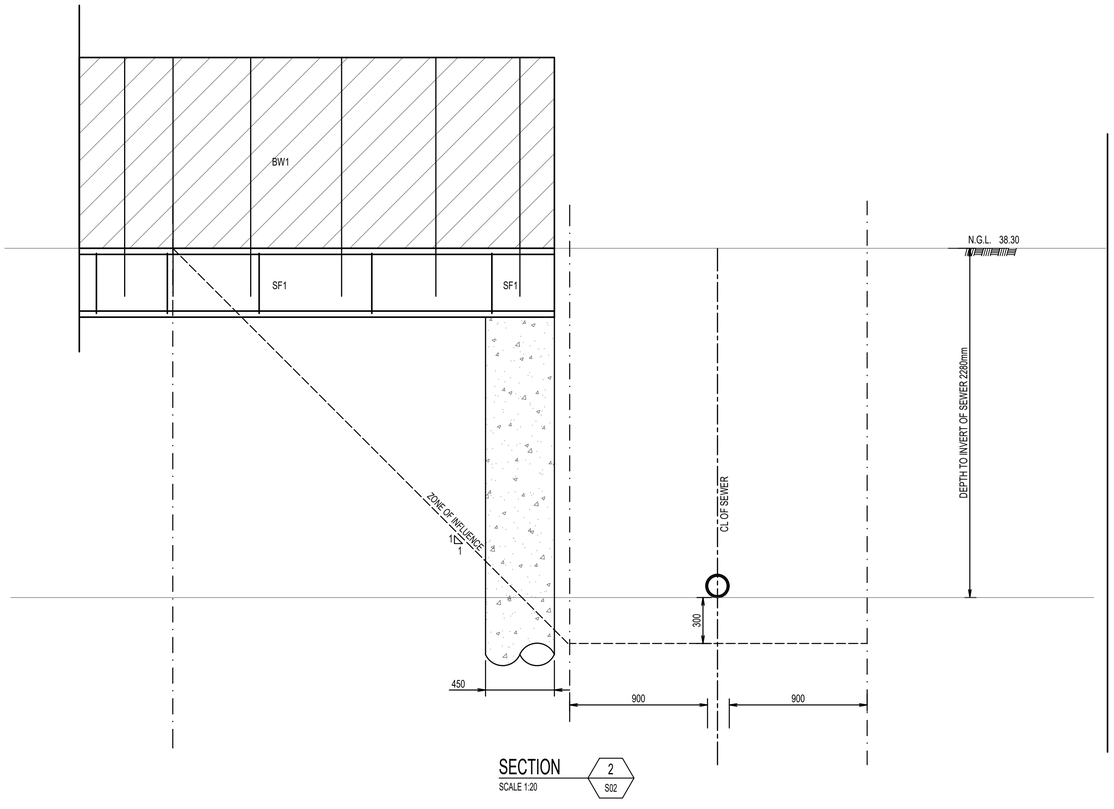
TYPICAL FENCE FOOTING FF

SCALE 1:20



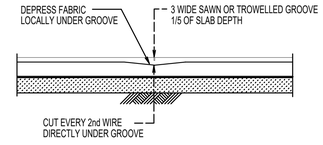
TYPICAL SLAB ON GROUND TRIMMER DETAILS

AT ALL COLUMNS, WALLS, PITS, FLOOR WASTES, ETC
THAT CAUSE A PENETRATION THROUGH THE SLAB.



SECTION 2 / S02

SCALE 1:20



S.J. - SAWN CONTROL JOINT

SCALE 1:20

NOTE: SAWCUT TO BE MADE WITHIN 16 HOURS OF CASTING SLAB

REV.	DATE	AMENDMENT DESCRIPTION	DRAWN
B	08.09.2026	FOR APPROVAL	TA
A	28.03.2026	FOR APPROVAL	TA

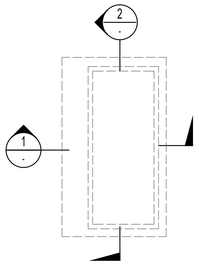
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PROPOSED DUAL-OCCUPANCY

18 CARRAMAR CRESCENT, MIRANDA NSW

GROUND FLOOR SECTIONS AND DETAILS

DRAWN	CHECKED	SCALE	DATUM	DRAWING No.	REV.
TA	TONY AHAL - Civil & Structural Engineer B.E. Civil (UTas) (MIEAust) 1060225	AS SHOWN	N/A	S018C - S02A	B



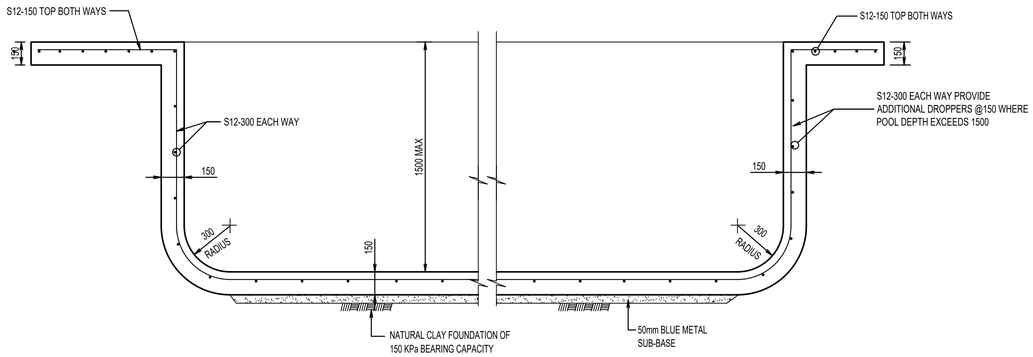
POOL SPECIFICATION

1. PLUMING IS TO BE IN ACCORDANCE WITH WRITTEN RECOMMENDATIONS OF FILTER MANUFACTURER
2. WALKWAYS ARE NOT DESIGNED TO SUPPORT MASONARY WALLS UNLESS OTHERWISE NOTED
3. DESIGN LIVE LOAD FOR WALKWAYS AND CONCOURSES 3 kPa

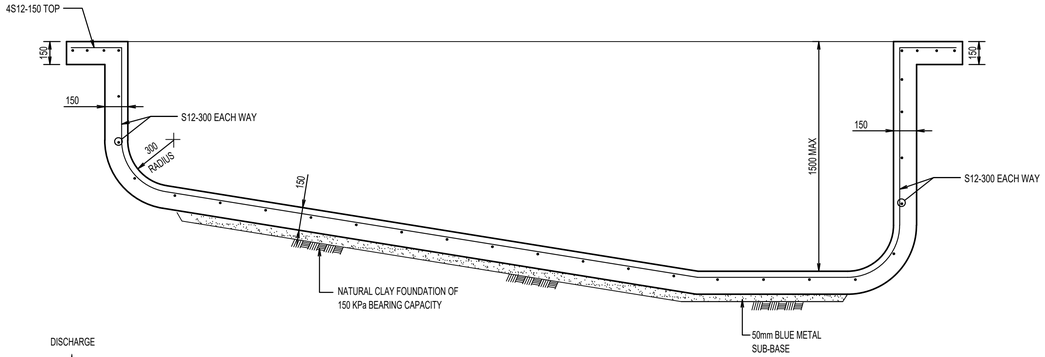
CONSTRUCTION NOTES

1. ALL DIMENSIONS LOCATING POOL ARE BE TAKEN FROM ARCHITECTURAL DRAWINGS
2. WRITTEN DIMENSIONS TO BE TAKEN IN REFERENCE TO SCALE
3. ENGINEER TO BE ADVISED IF EXCAVATION IS IN FILL OR IF EXCESSIVE GROUND WATER IS ENCOUNTER
4. SUPPORTING FOUNDATION MATERIAL TO BE STIFF CLAY OF UNIFORM MOISTURE CONTENT WITH SAFE BEARING CAPACITY OF 150 kPa
5. WHERE IT IS CONSIDERED THAT GROUND WATER CAN BUILD UP TO A LEVEL 500mm ABOVE THE FLOOR OF THE EXCAVATION ADEQUATE DRAINAGE SHALL BE PROVIDED UNDER THE POOL FLOOR
6. CONCRETE TO HAVE A MINIMUM DESIGN STRENGTH OF f'c = 25 MPa AT 28 DAYS CONCRETE TO BE PNEUMATICALLY APPLIED
7. UPON COMPLETION OF CONCRETING THE HYDROSTATIC VALVE IS TO BE CHECKED TO ENSURE EFFECTIVE AND SUFFICIENT OPERATION
8. ALL REINFORCEMENT TO BE SUPPORTED BY BAR CHAIRS
9. REINFORCEMENT IS TO BE STRUCTURAL GRADE DEFORMED BAR GRADE TO 230 TO AUSTRALIAN STANDARD AS1302
10. WATER FACE REINFORCEMENT TO HAVE 60mm CONCRETE COVER. REAR FACE REINFORCEMENT HAVE 50mm COVER FROM REAR FACE IF FORMED, AND 60mm COVER IF SPRAYED AGAINST GROUND
11. ALL BARS SHALL BE SPLICED 40 BAR DIAMETERS MIN.
12. SPLICES IN BOND BEAM BARS SHALL BE STAGGERED
13. ALTERNATIVE REINFORCEMENT TO BE TEMCORE BARS IN ACCORDANCE WITH AS 1302 410 Y

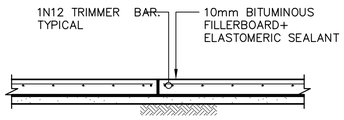
CONCRETE QUALITY				
ELEMENT	SLUMP	AGGREGATE (MAX. SIZE)	CEMENT TYPE	f'c
POOL BASE / WALLS	80mm	20mm	A	32 MPa



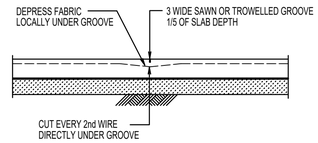
SECTION 1
SCALE 1:20



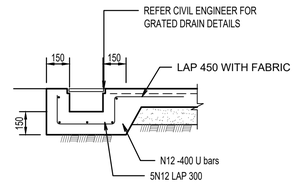
SECTION 2
SCALE 1:20



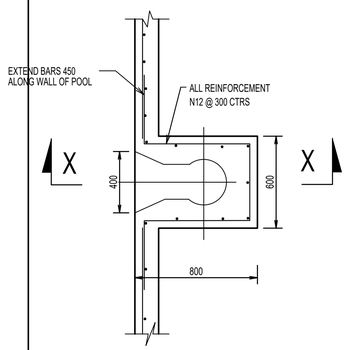
BUTT JOINT 'BJ'



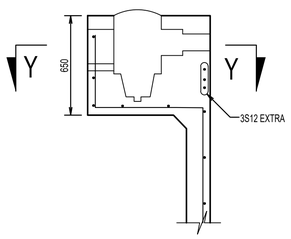
S.C. - SAWN CONTROL JOINT
SCALE 1:20
NOTE: SAWCUT TO BE MADE WITHIN 16 HOURS OF CASTING SLAB



TYPICAL GRATED DRAIN DETAIL
SCALE 1:20

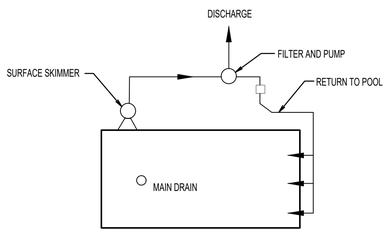


PLAN - SECTION Y-Y

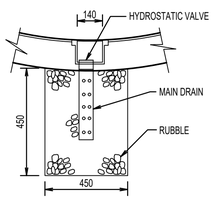


ELEVATION - SECTION X-X

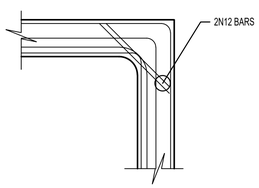
SKIMMER BOX DETAIL
SCALE 1:20



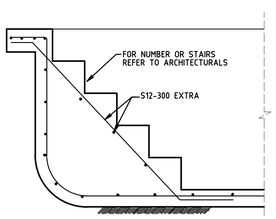
PLUMBING LAYOUT
DIAGRAMATIC ONLY



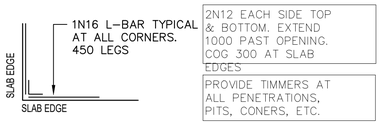
MAIN DRAIN DETAIL
SCALE 1:20

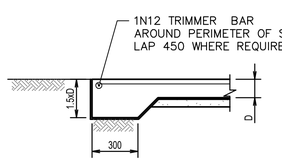


BOND BEAM CORNER DETAIL
SCALE 1:50

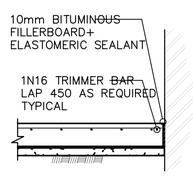


TYPICAL STAIR SECTION





SLAB ON GROUND EDGE THICKENING DETAILS



BUTT JOINT TO WALL

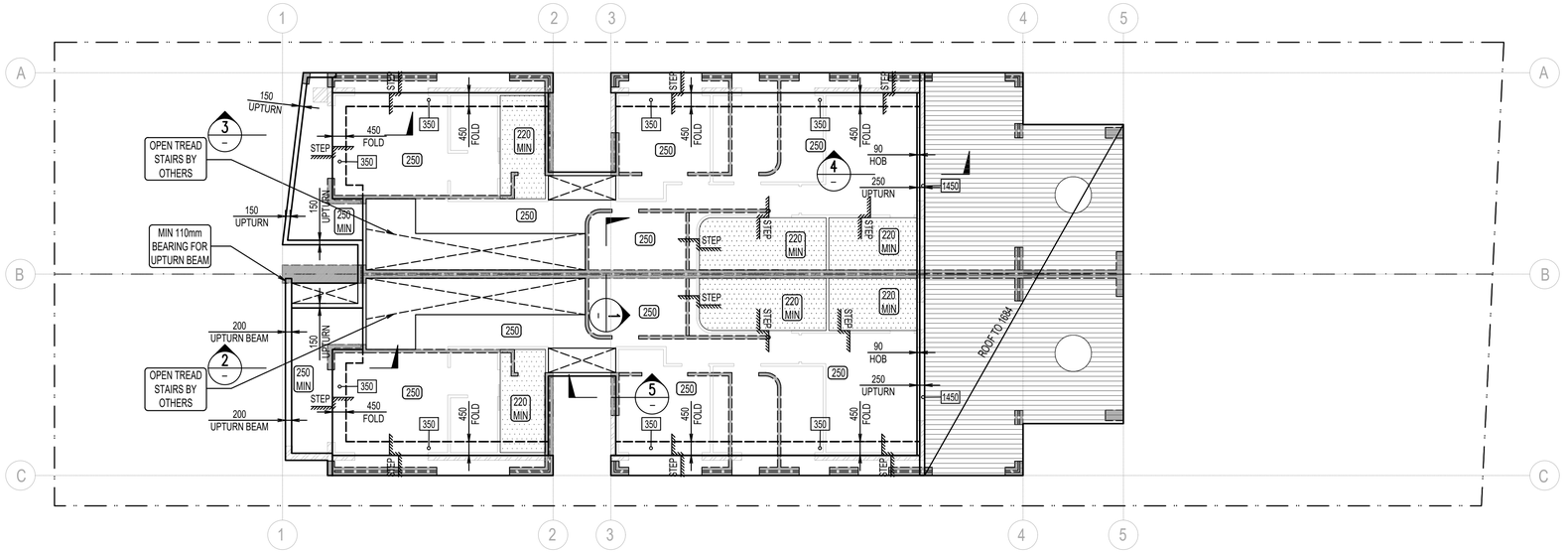
REV.	DATE	AMENDMENT DESCRIPTION	DRAWN
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PROPOSED DUAL-OCCUPANCY
18 CARRAMAR CRESCENT, MIRANDA NSW

SWIMMING POOL SECTIONS AND DETAILS

DRAWN	CHECKED	SCALE	DATUM	DRAWING No.	REV.
TA	TONY AHAL - Civil & Structural Engineer B.E. Civil (USyd) (MIEAust) 1080225	AS SHOWN	N/A	S018C - S02B	B



FIRST FLOOR PLAN

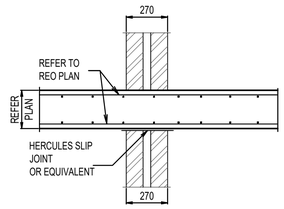
SCALE 1:100

NOTE :

1. ALL EXPOSED SLABS TO BE WATERPROOFED WITH APPROVED SYSTEM.
2. CHAMFER, FILLET AND DRIP GROOVES TO ARCHT SPECIFICATIONS AND TO BUILDER REQUIREMENTS.
3. ALL BRICKS WALLS TO BE ARTICULATED AND SPACED IN ACCORDANCE WITH AS3700 AND ARTICULATED WALLING GUIDELINE TN61
4. ACOUSTIC ENGINEER TO CHECK AND VERIFY ALL FLOOR SLABS FOR ACOUSTIC REQUIREMENTS IN ACCORDANCE WITH THE BCA REGULATIONS AND ADVISE ON ANY ACOUSTIC TREATMENT NEEDED TO SATISFY THE REGULATIONS.
5. DRAWING TO BE READ IN CONJUNCTION WITH ARCHITECTURALS.
6. REFER TO ARCHITECTURAL DRAWINGS FOR ALL SETOUT, LEVELS, FALLS ETC.
7. ALL STAIRS TO ARCHITECTURAL SPECIFICATIONS

CONCRETE QUALITY				
ELEMENT	SLUMP	AGGREGATE (MAX. SIZE)	CEMENT TYPE	f'c
SUSPENDED SLAB	80mm	20mm	A	40 MPa
BEAMS	80mm	20mm	A	40 MPa

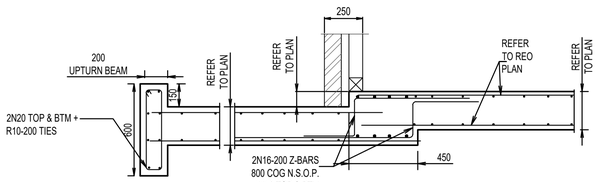
REINFORCEMENT COVER SCHEDULE				
MEMBER	COVER (mm)			EXPOSURE CLASSIFICATION
	TOP	BOTTOM	SIDES	
INTERNAL SLAB	30mm	30mm	30mm	A1
EXTERNAL SLAB	40mm	40mm	40mm	A2
BEAMS	40mm	40mm	40mm	A1



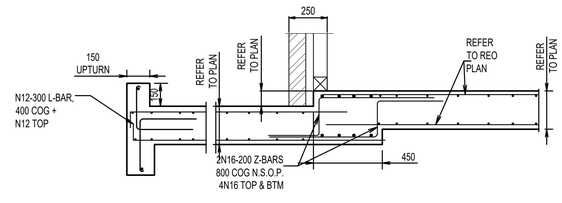
SECTION 1
SCALE 1:20

LEGEND	
	LOAD BEARING BRICK WALL BELOW
	LOAD BEARING BRICK WALL ABOVE
	30mm WET AREA SET-DOWNS
	DENOTES SLAB VOID
200	DENOTES SLAB THICKNESS
	DENOTES STEP IN SLAB

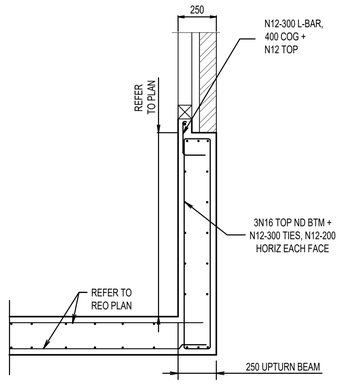
FIRST FLOOR WALLS TO BE BRICK VENEER ENGINEER TO BE NOTIFIED IF OTHERWISE



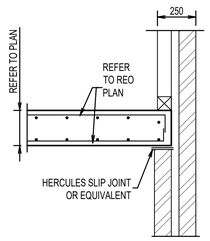
SECTION 2
SCALE 1:20



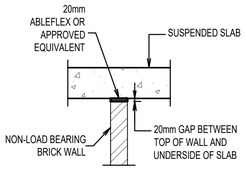
SECTION 3
SCALE 1:20



SECTION 4
SCALE 1:20

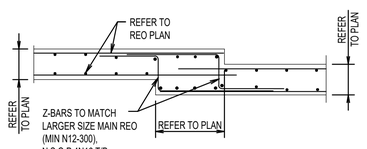


SECTION 5
SCALE 1:20



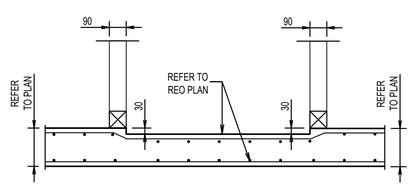
NON-LOAD BEARING BRICK WALL DETAIL

SCALE 1:20



TYPICAL STEP DETAIL

SCALE 1:20



TYPICAL SETDOWN DETAIL

SCALE 1:20

REV.	DATE	AMENDMENT DESCRIPTION	DRAWN
B	08.09.2026	FOR APPROVAL	TA
A	28.03.2026	FOR APPROVAL	TA

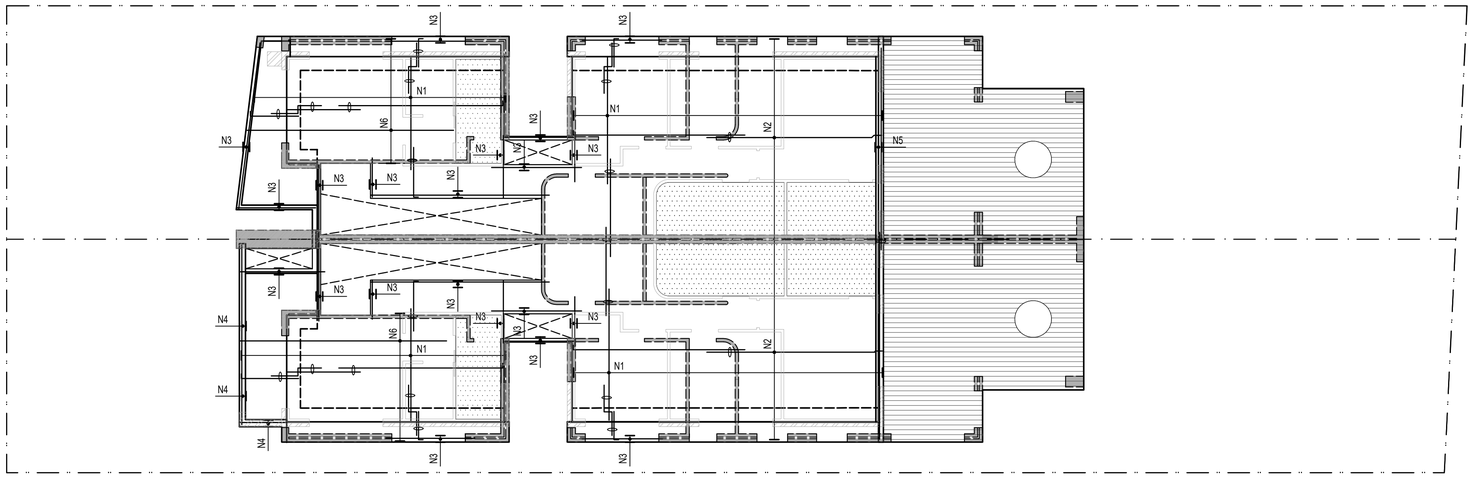
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PROPOSED DUAL-OCCUPANCY

18 CARRAMAR CRESCENT, MIRANDA NSW

FIRST FLOOR SLAB PLAN, SECTIONS AND DETAILS

DRAWN	CHECKED	SCALE	DATUM	DRAWING No.	REV.
TA	TONY AHAL - Civil & Structural Engineer B.E. Civil (USyd) (MIEAust) 1080225	AS SHOWN	N/A	S018C - S03	B



BOTTOM REINFORCEMENT PLAN

FRST FLOOR REINFORCEMENT PLAN

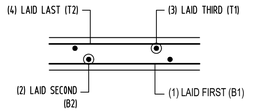
SCALE 1:100 A1 1:200 A3
NOTE :

BOTTOM REINFORCEMENT	TOP REINFORCEMENT
N1 - N12-200 (1)	N7 - N12-200 (3)
N2 - N12-200 (2)	N8 - N12-200 (4)
N3 - 3N16 EXTRA	N9 - 3N16 EXTRA
N4 - NOT USED	N10 - 3N20 EXTRA
N5 - 3N16 + N12-300 TIES	N11 - NOT USED
N6 - N16-400 EXTRA	N12 - 3N16 + N12-300 TIES
	N13 - N16-400 EXTRA

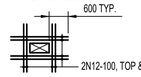
LEVEL 1 REINFORCEMENT NOTES:

1. ALL REINFORCEMENT SHOWN IS TO BE ADDITIONAL TO STANDARD REINFORCEMENT. REFER TO SECTIONS ON THIS DRAWING FOR STANDARD REINFORCEMENT.

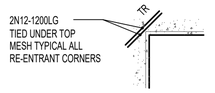
2. SLAB REINFORCEMENT LAYING SEQUENCE U.N.O.:



3. TYPICAL TRIMMERS AT PENETRATION DETAIL:



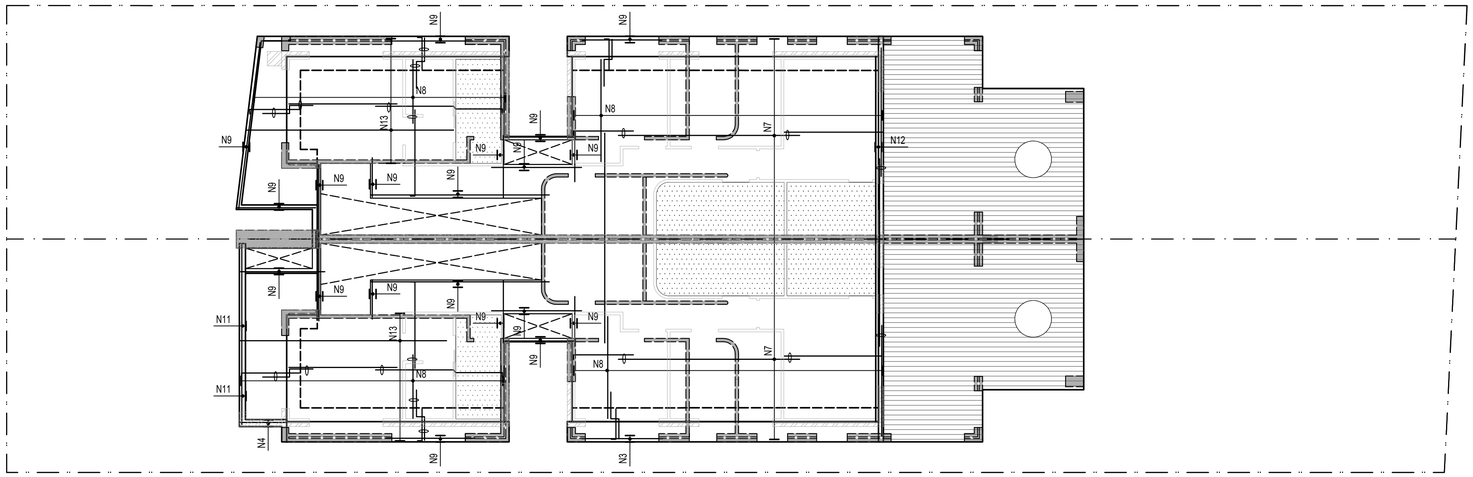
4. TYPICAL TRIMMERS AT RE-ENTRANT CORNERS



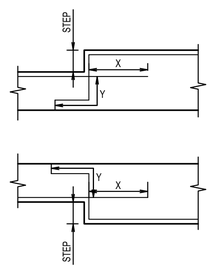
5. PROVIDE TOP & BOTTOM BARS ON ALL WALL OPENINGS AS PER THE SCHEDULE BELOW:

LENGTH OF OPENING (m)	BOTTOM BAR	TOP BAR
1.0 - 1.8	2N12	2N12
1.8 - 3.5	2N16	2N16
3.5 - 4.5	2N20	2N20

6. PROVIDE ADEQUATE OVERLAP AT MESH JOINTS AS FOLLOWS :
400 MIN. LAP FOR SL82 MESH
500 MIN. LAP FOR SL92 MESH
600 MIN. LAP FOR SL102 MESH



TOP REINFORCEMENT PLAN



PROVIDE ADEQUATE OVERLAP FOR BARS AS FOLLOWS :

BAR	MIN LAPPING REQUIRED (X-Y)
N12	500 mm
N16	800 mm
N20	1000 mm
N24	1200 mm
N28	1400 mm

TYPICAL REINFORCEMENT LAPPING

REV.	DATE	AMENDMENT DESCRIPTION	DRAWN
B	08.09.2026	FOR APPROVAL	TA
A	28.03.2026	FOR APPROVAL	TA

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PROPOSED DUAL-OCCUPANCY
18 CARRAMAR CRESCENT, MIRANDA NSW

FIRST FLOOR BTM AND TOP REO PLAN

DRAWN	CHECKED	SCALE	DATUM	DRAWING No.	REV.
TA	TONY AHAL - Civil & Structural Engineer B.E. Civil (USyd) (MIEAust) 1080225	AS SHOWN	N/A	S018C - S03A	B